

Microsoft Sentinel

SOC Monitoring & Incident Investigation Using Microsoft Sentinel

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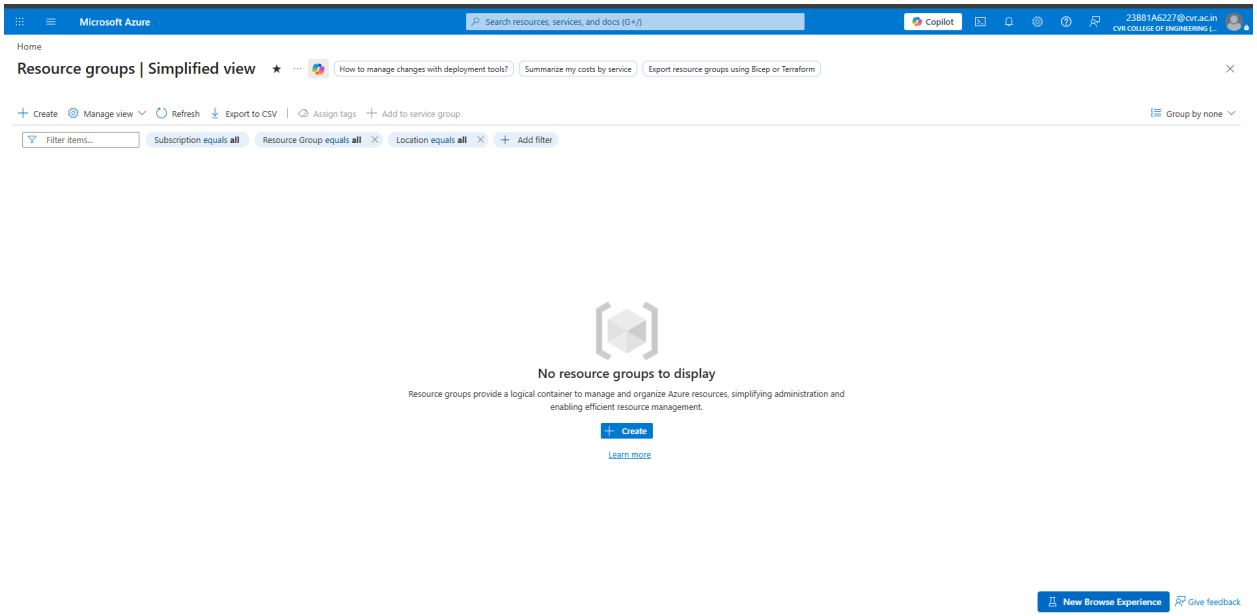
2026–2027

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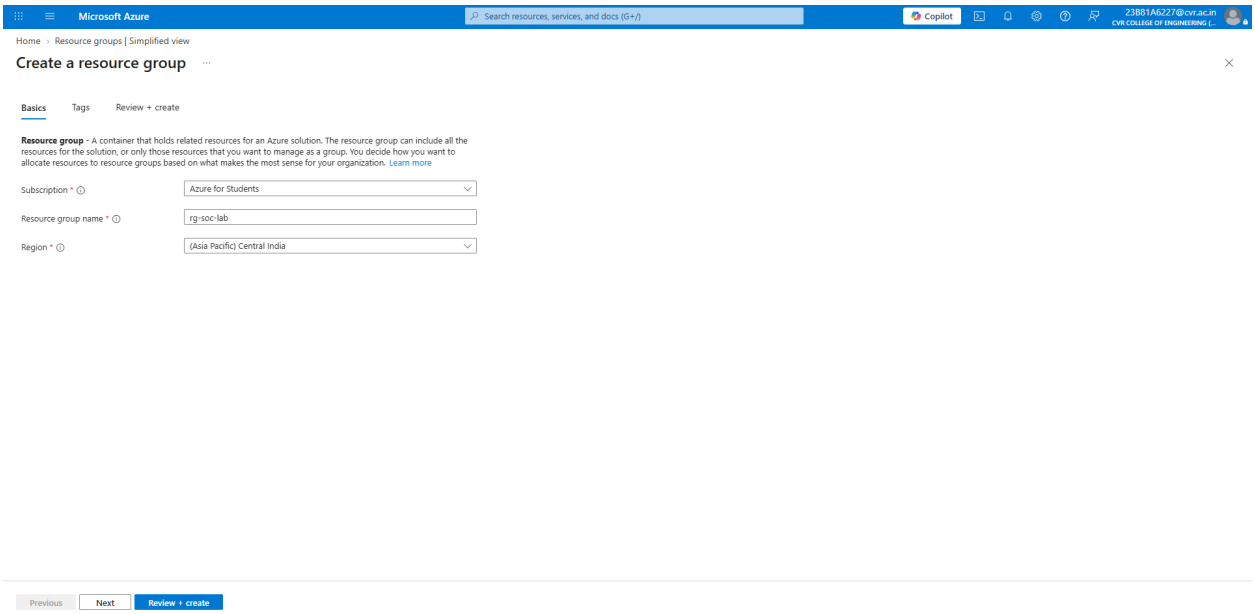
I. Create a Resource Group

1. Type resource group in search bar and click **create**.

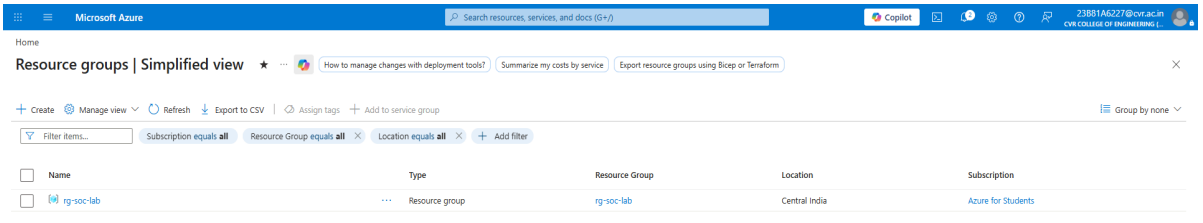


2. Resource group name: **rg-soc-lab**

   Region: **Central India**

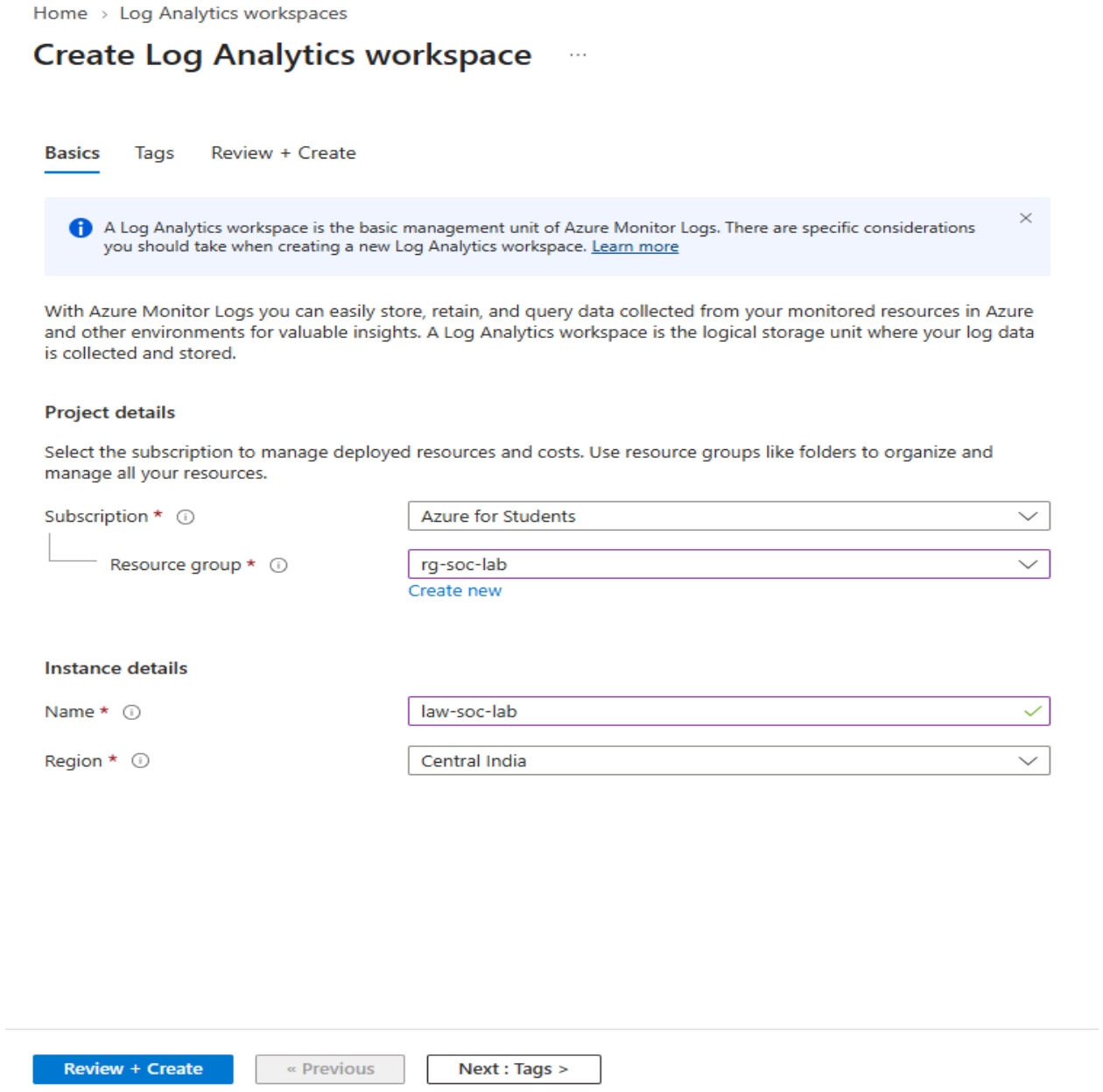


   Then click **Create.** The resource group is created and shown like this.

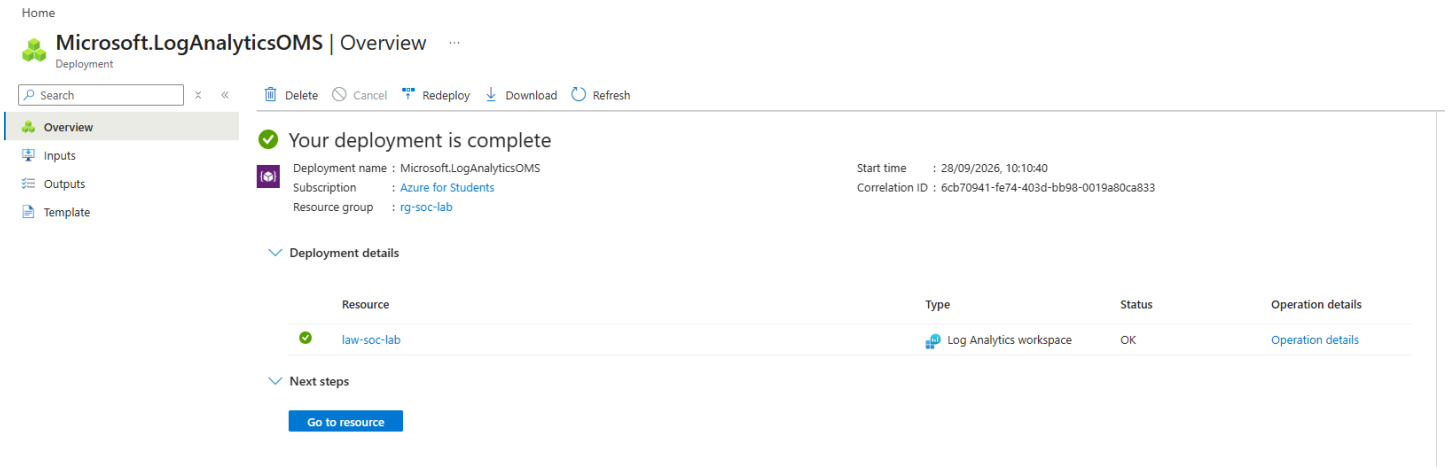


II. Log Analytics Workspace

1. Search `Log Analytics workspaces` → **Create**
2. Enter details as shown. Then click **Review + Create.** Then click **Create.**

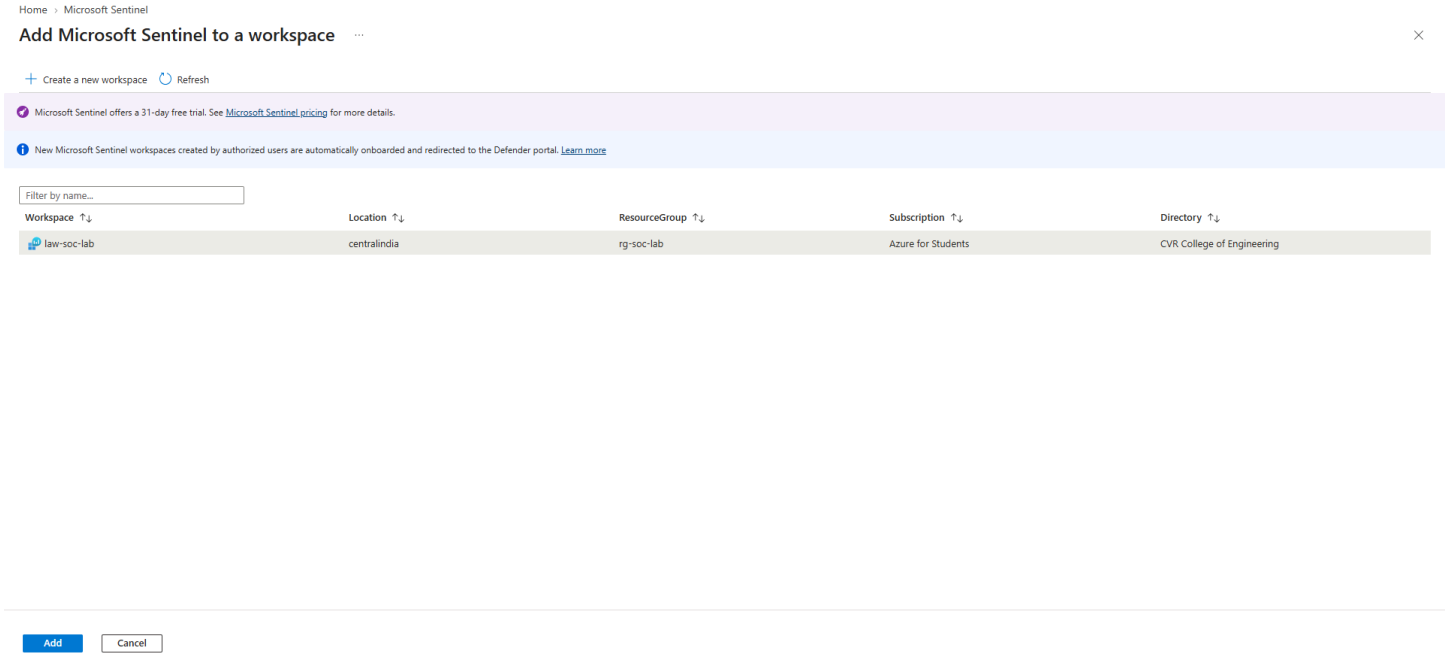


3. Then it is created like this.



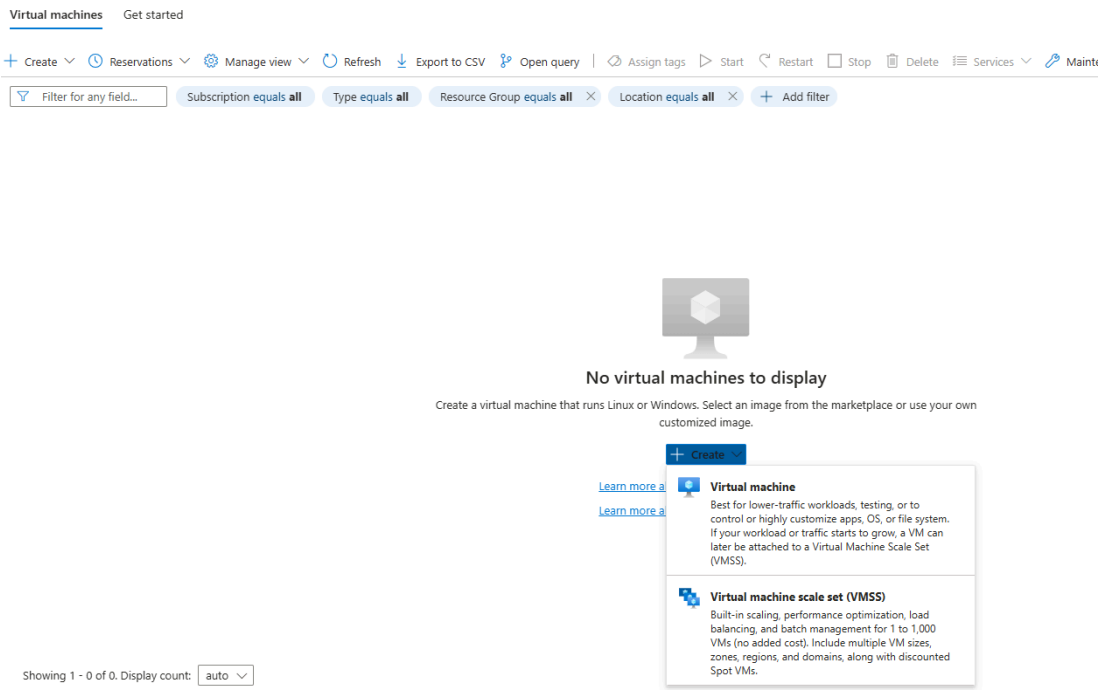
III. Microsoft Sentinel

1. In the **Azure Portal**, search for **Microsoft Sentinel** and open it.
2. Select your workspace (**law-soc-lab**). Click on **Add.**



IV. Create the Windows VM

1. Click on Policy → Assignment → Allowed Resource deployment regions. There we find ["indiasouthcentral","malaysiawest","centralindia","koreacentral","eastasia"]. We will use India South Central.
2. Search for **Virtual machines**. Then click **Create → Virtual machine.**



3. Enter these details.

Setting	Select
Subscription	Your subscription
Resource group	`rg-soc-lab`
VM name	`MyWindowsVM`
Region	IndiaSouthCentral
Image	**Windows 11 Pro**
Size	We'll choose a small eligible size
Username	`azureuser`
Password	<your_password>

Create a virtual machine ...

Basics Disks Networking Management Monitoring Advanced Tags **Review + create**

View automation template

You have set RDP port(s) open to the internet. This is only recommended for testing. If you want to change this setting, go back to Basics tab.

Basics

Subscription	Azure for Students
Resource group	rg-soc-lab
Virtual machine name	MyWindowsVM
Region	India South Central
Availability options	Availability zone
Availability zone	1
Zone options	Self-selected zone
Security type	Trusted launch virtual machines
Enable secure boot	Yes
Enable vTPM	Yes
Integrity monitoring	No
Image	Windows 11 Enterprise, version 25H2
Size	Standard B2ats_v2
VM architecture	x64
Username	azureuser
Public inbound ports	RDP (3389)
Already have a Windows license?	Yes
License type	Windows Client
Azure Spot	No

Home > Compute infrastructure | Virtual machines

Create a virtual machine ...

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags

Disks

OS disk size	Image default
OS disk type	Premium_LRS
Use managed disks	Yes
Delete OS disk with VM	Yes
Ephemeral OS disk	None

Networking

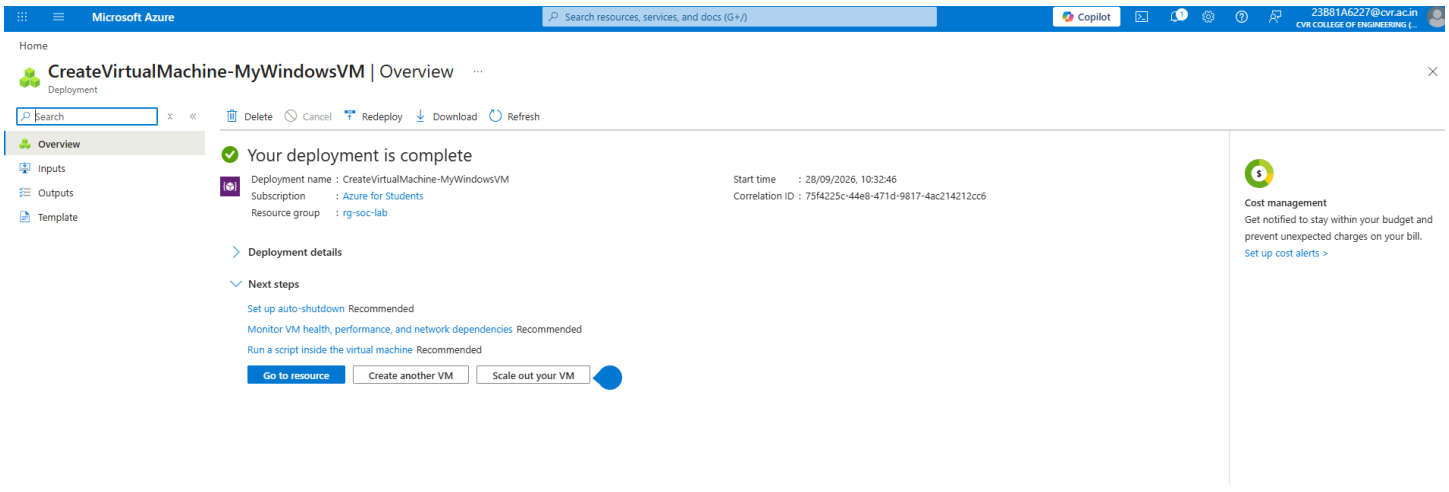
Virtual network	(new) vnet-indiasouthcentral-1
Subnet	(new) snet-indiasouthcentral-1
Public IP	(new) MyWindowsVM-ip
NIC Network security group	-
Accelerated networking	Off
Place this virtual machine behind an existing load balancing solution?	No
Delete public IP and NIC when VM is deleted	Disabled

Management

System assigned managed identity	Off
Login with Microsoft Entra ID	Off
Backup	Off
Site Recovery	Off
Enable periodic assessment	Off
Enable hotpatch	Off
Patch orchestration options	Automatic by OS (Windows Automatic Updates)
Auto-shutdown	Off
Enable hibernation	Off

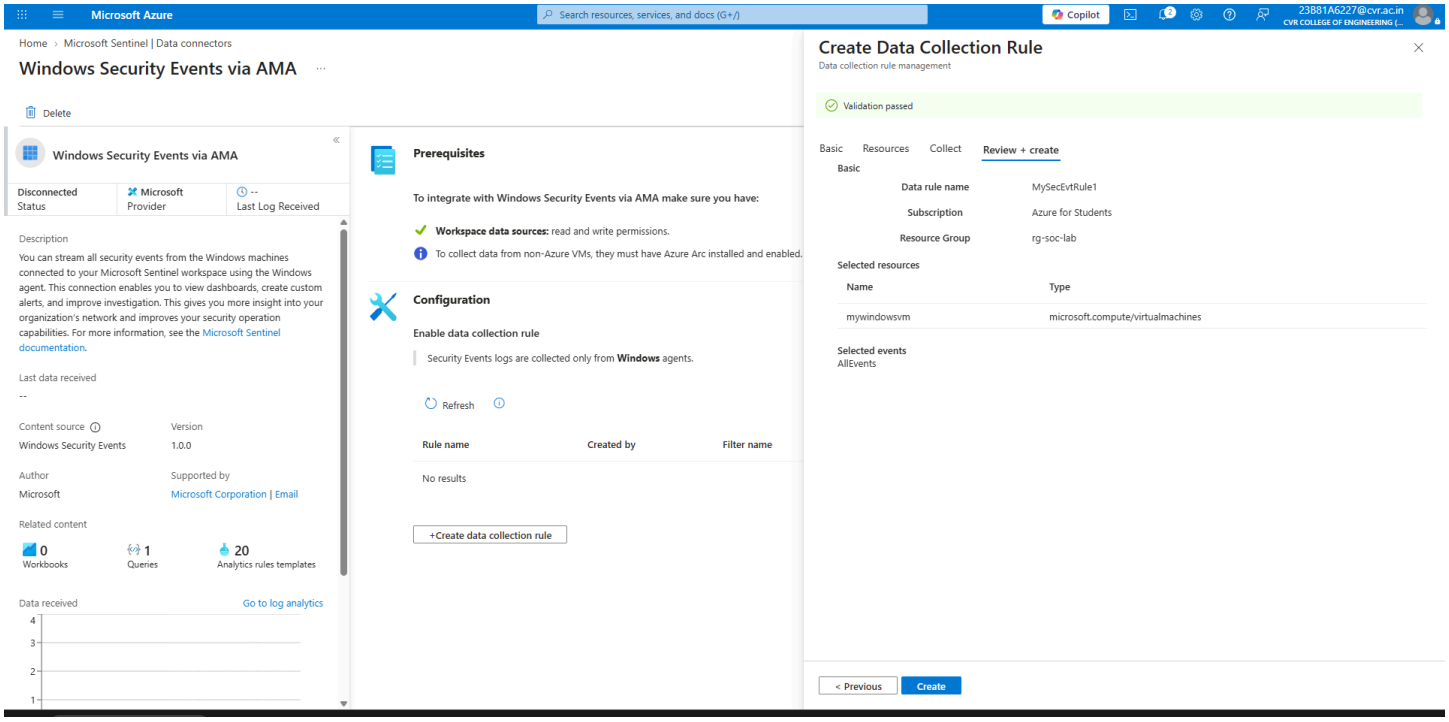
Previous | Next | Create

4. After clicking create, we will get this.

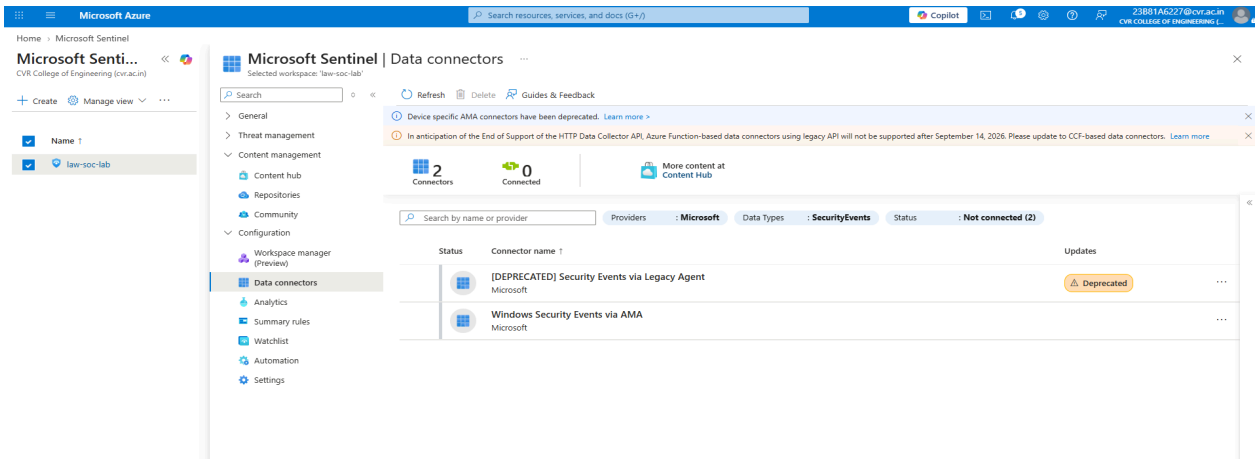


V. Microsoft Sentinel Data Connector

1. Search Microsoft Sentinel. Then open **law-soc-lab**
2. Click **Configuration → Data connectors**
3. Search for: **Windows Security Events** via AMA. Then click **connector** page
4. Click: **+ Create data collection rule.** This opens the panel on the right.
5. Enter:
    - Data rule name: `MySecEvtRule1`
    - Subscription: `Azure for Students`
    - Resource group: `rg-soc-lab`
6. Click Next. Click + Add resource(s). Select `MyWindowsVM`, then click Apply.
7. Click Next. Select **Common.**
8. Click **Review + create.** Make sure Validation passed is displayed.
9. Click **Create.**

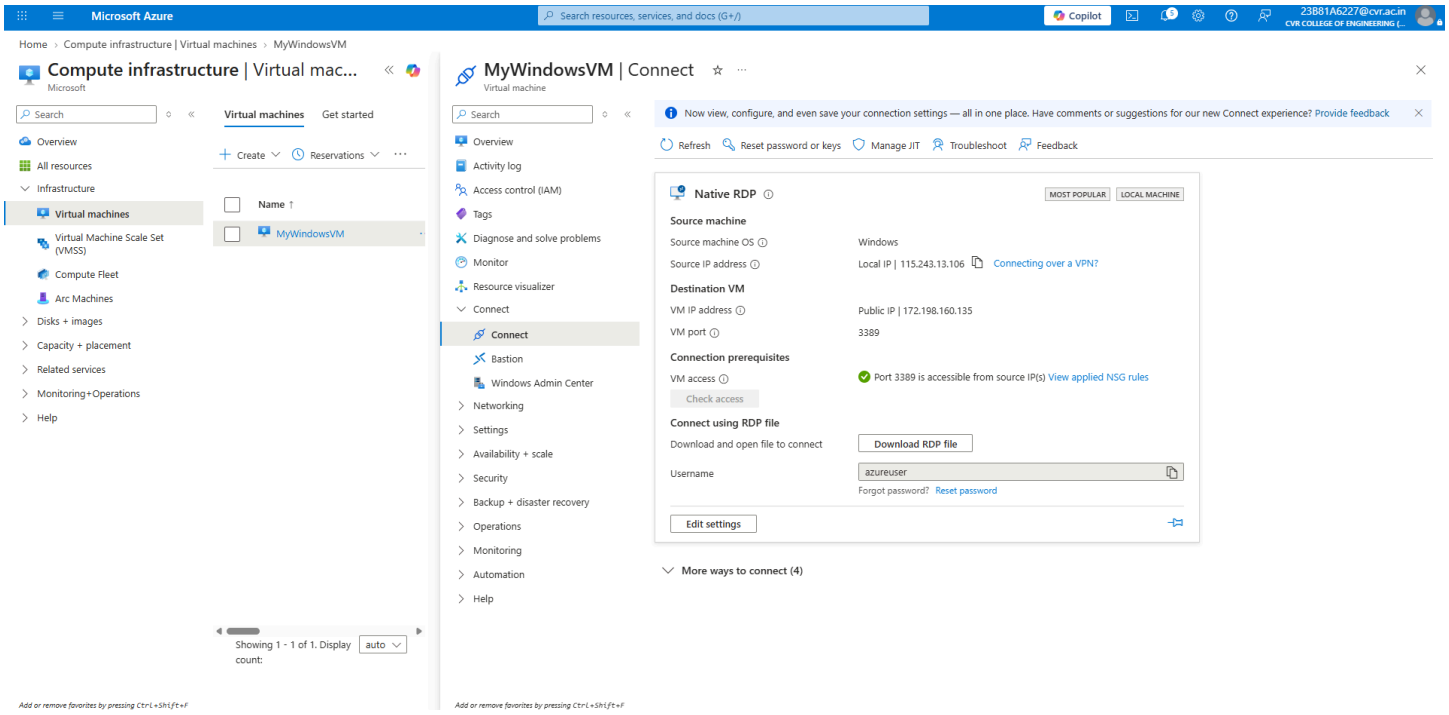


10. Return to the Windows Security Events via AMA connector page. Click Refresh and verify that the Data Collection Rule appears under Configuration.

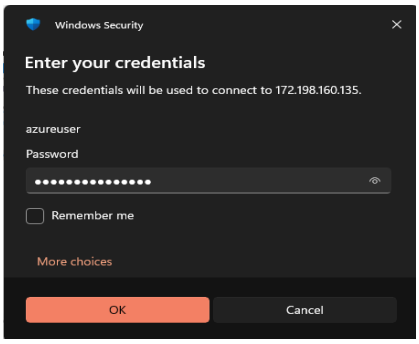


VI. Generate Windows security events and verify that Sentinel is receiving them.

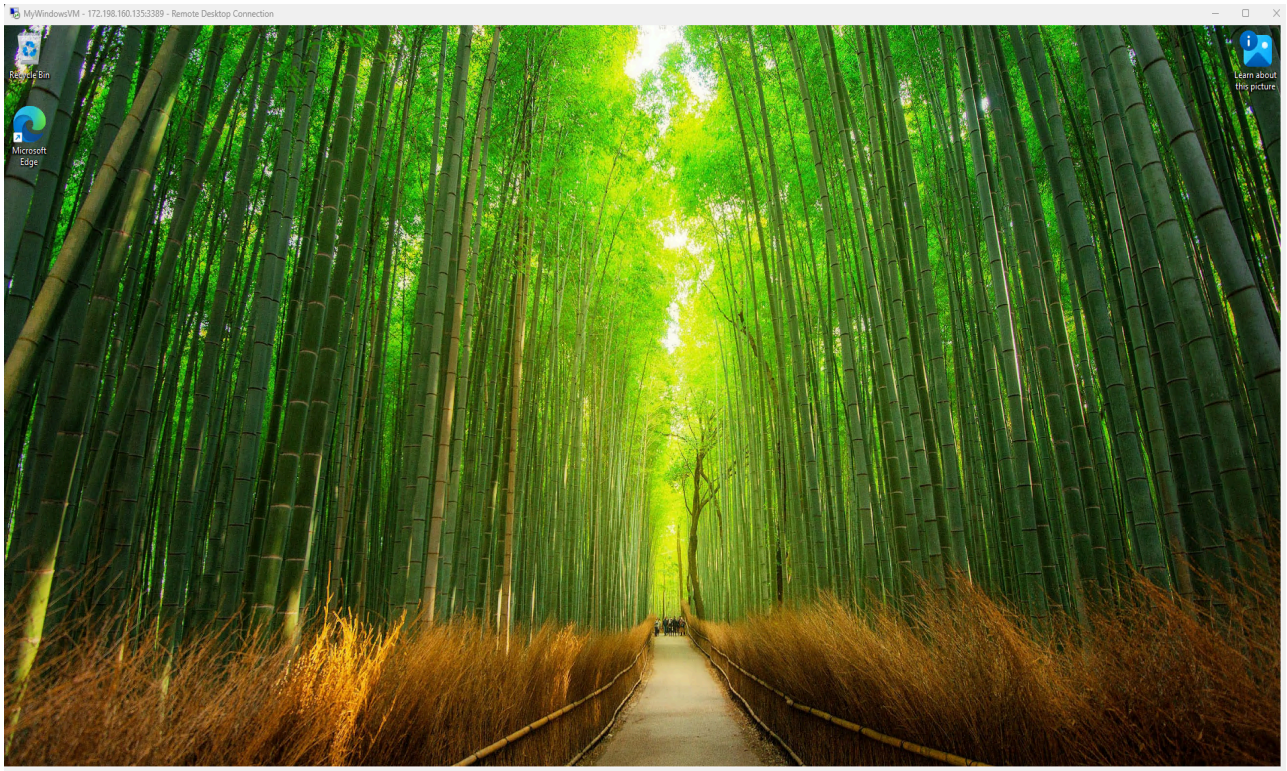
1. Go to Virtual Machines.
2. Open **MyWindowsVM → Connect → RDP → Download RDP file**.



3. Open the downloaded `.rdp` file on your PC and connect using the **username and password** you created for the VM.



4. Then we get connected to **azureuser** remotely.

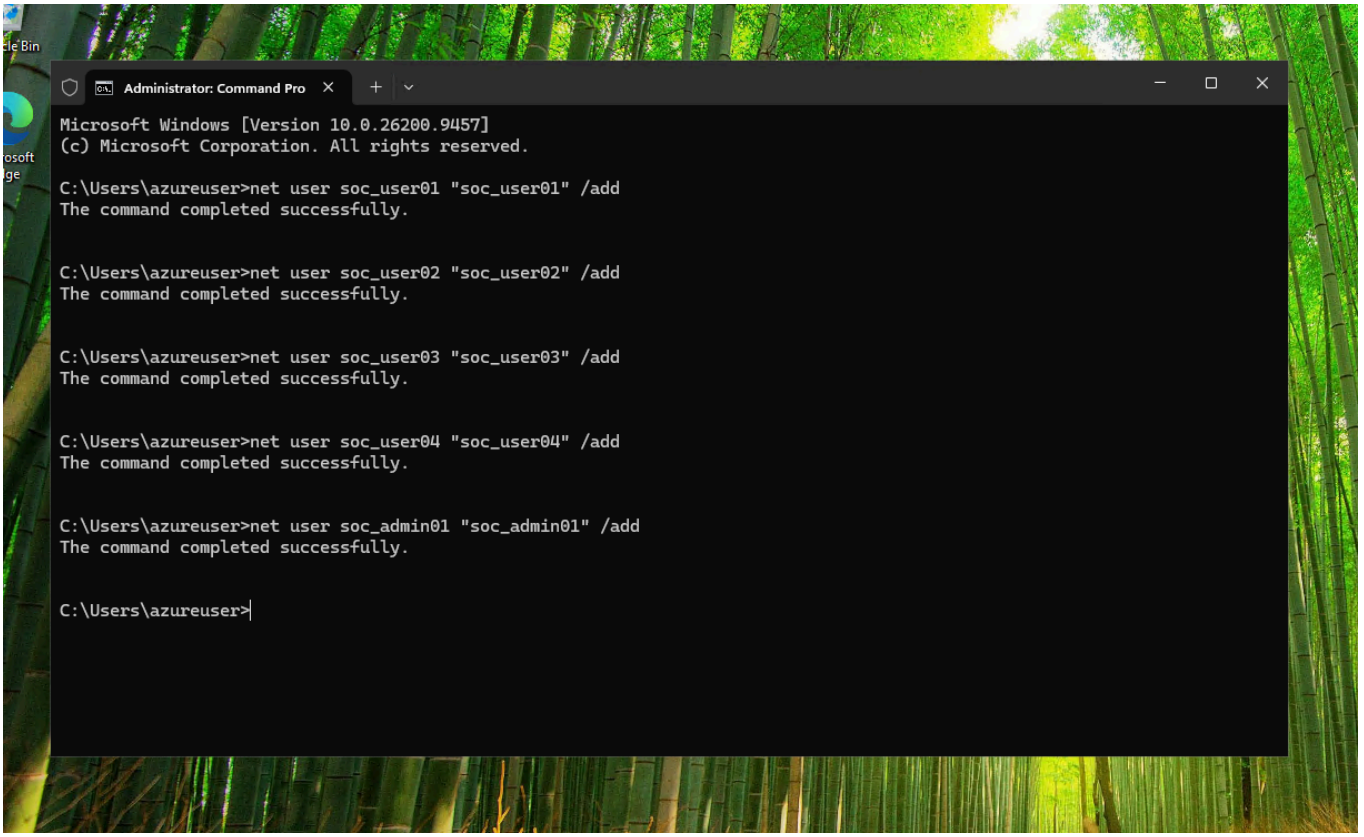


(or)

- Install **Windows App** (if you dont want to download and use the RDP file to access the Windows VM)
- If you face errors while connecting:

  Go to: **Azure Portal → MyWindowsVM → Overview.** Then Click **Restart.**

5. Open **cmd, run as administrator.**
   Enter the following commands, to create users and an admin. (We created 4 soc users and 1 soc admin user).
   Syntax: **net user <username> "<password>" /add**



6. Check that the users were created using **net user** command.

```
C:\Users\azureuser>net user

User accounts for \\MyWindowsVM

-------------------------------------------------------------------------------
azureuser                DefaultAccount           Guest
soc_admin01              soc_user01               soc_user02
soc_user03               soc_user04               WDAGUtilityAccount
The command completed successfully.
```

- **net user** allows you to check the existing user.
- You can also check an individual user. Ex: **net user soc_user01.**

```
C:\Users\azureuser>net user soc_user01
User name                    soc_user01
Full Name
Comment
User's comment
Country/region code          000 (System Default)
Account active               Yes
Account expires              Never

Password last set            9/28/2026 6:30:56 AM
Password expires             11/9/2026 6:30:56 AM
Password changeable          9/28/2026 6:30:56 AM
Password required            Yes
User may change password     Yes

Workstations allowed         All
Logon script
User profile
Home directory
Last logon                   Never

Logon hours allowed          All

Local Group Memberships      *Users
Global Group memberships     *None
The command completed successfully.
```

7. **net localgroup** command allows you to see all the local groups available on the Windows VM.

```
C:\Users\azureuser>net localgroup

Aliases for \\MyWindowsVM

-------------------------------------
*Access Control Assistance Operators
*Administrators
*Backup Operators
*Cryptographic Operators
*Device Owners
*Distributed COM Users
*Event Log Readers
*Guests
*Hyper-V Administrators
*IIS_IUSRS
*Network Configuration Operators
*OpenSSH Users
*Performance Log Users
*Performance Monitor Users
*Power Users
*Remote Desktop Users
*Remote Management Users
*Replicator
*System Managed Accounts Group
*User Mode Hardware Operators
*Users
The command completed successfully.
```

8. Add `soc_admin01` to the Administrators group using the command: **net localgroup Administrators soc_admin01 /add.** This gives `soc_admin01` local administrator privileges.

```
C:\Users\azureuser>net localgroup Administrators soc_admin01 /add
The command completed successfully.
```

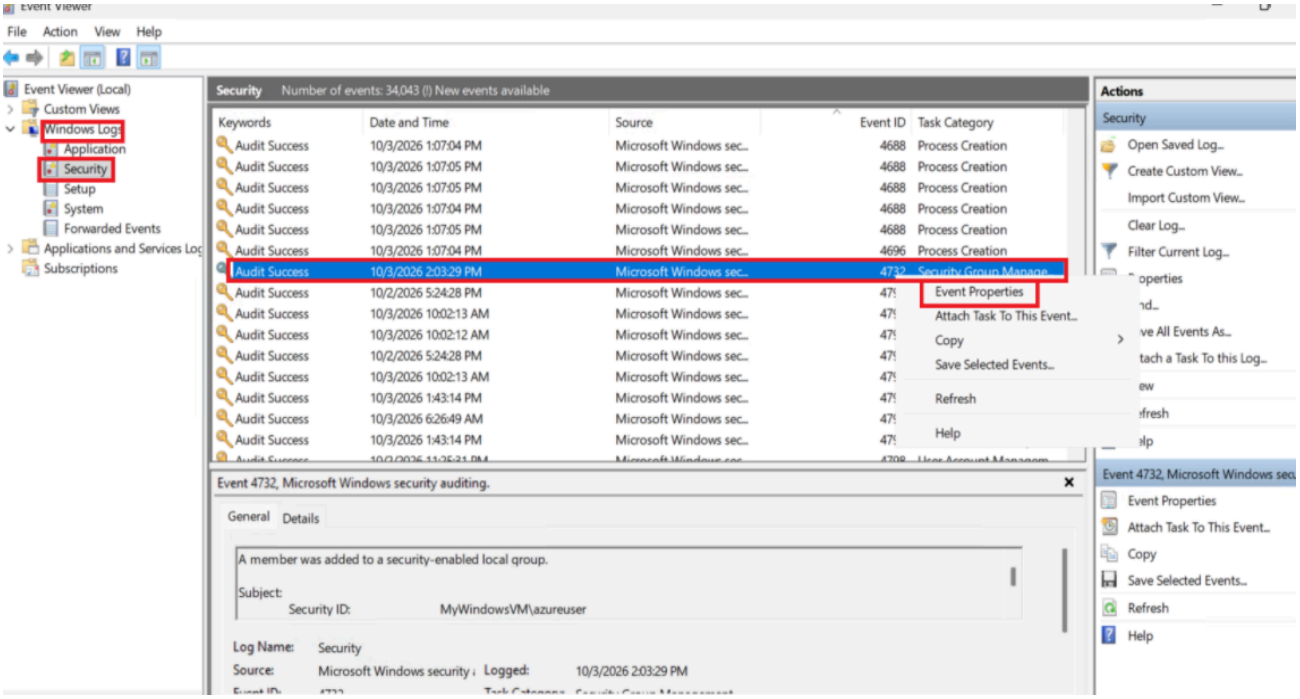
9. Verify that `soc_admin01` was added successfully using the command**: net localgroup Administrators.** The `soc_admin01` account should now appear in the list.

```
C:\Users\azureuser>net localgroup Administrators
Alias name     Administrators
Comment        Administrators have complete and unrestricted access to the computer/domain

Members

-------------------------------------------------------------------------------
azureuser
soc_admin01
The command completed successfully.
```

10. Open **Event Viewer** by searching for it in the Windows search bar.
11. Go to **Windows Logs** → **Security**.
12. Look for **Event ID 4732**. This event is generated when a member is added to a security-enabled local group.
13. Right-click the **4732** event and select **Event Properties** to view the details of the event.





14. Open Command Prompt as Administrator and add `soc_user01` to the Remote Desktop Users group: **net localgroup "Remote Desktop Users" soc_user01 /add.** Add all the users similarly including soc_admin01.

```
C:\Users\azureuser>net localgroup "Remote Desktop Users"  soc_user01 /add
The command completed successfully.

C:\Users\azureuser>net localgroup "Remote Desktop Users"  soc_user02 /add
The command completed successfully.

C:\Users\azureuser>net localgroup "Remote Desktop Users"  soc_user03 /add
The command completed successfully.

C:\Users\azureuser>net localgroup "Remote Desktop Users"  soc_user04 /add
The command completed successfully.

C:\Users\azureuser>net localgroup "Remote Desktop Users"  soc_admin01 /add
The command completed successfully.
```

This adds all the users to the **Remote Desktop Users** group, allowing the account to connect to the Windows VM through Remote Desktop.

15. Verify that the users were added successfully: **net localgroup "Remote Desktop Users"**

```
C:\Users\azureuser>net localgroup "Remote Desktop Users"
Alias name     Remote Desktop Users
Comment        Members in this group are granted the right to logon remotely

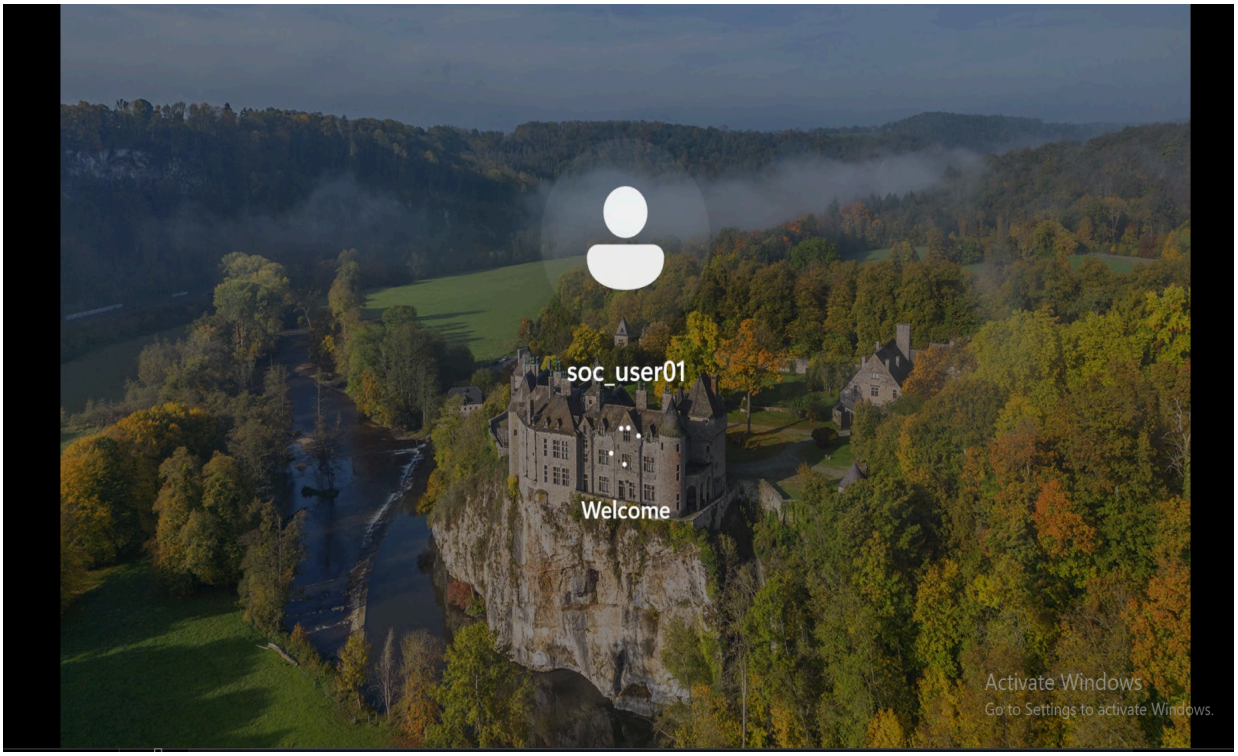
Members

-------------------------------------------------------------------------------
soc_admin01
soc_user01
soc_user02
soc_user03
soc_user04
The command completed successfully.
```

16. Close the current **Remote Desktop** session.
17. Open **Remote Desktop / Windows App**

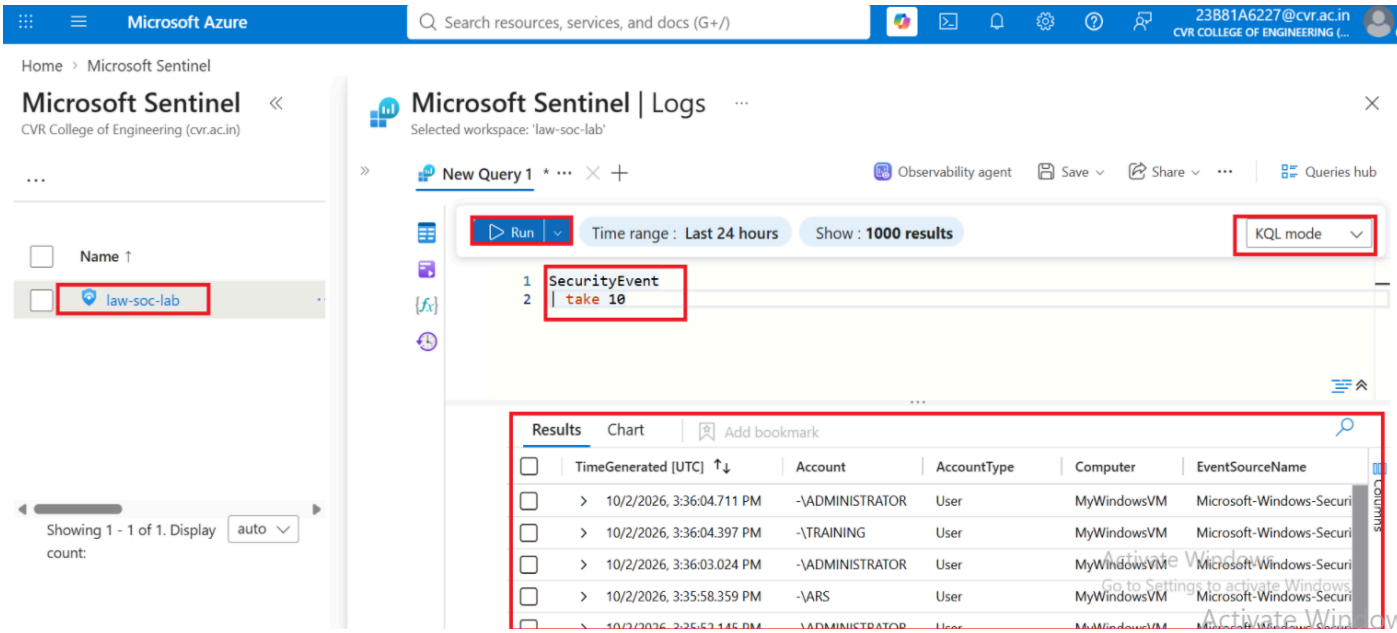
18. When prompted for credentials, select **Use another account** and enter:

    Username: `soc_user01`
    Password: `soc_user01`



19. After successfully connecting, open **Microsoft Sentinel → law-soc-lab → Logs**.
20. Switch from Simple Mode to **KQL Mode.**
21. Run the following query to verify that Windows Security events are being received:

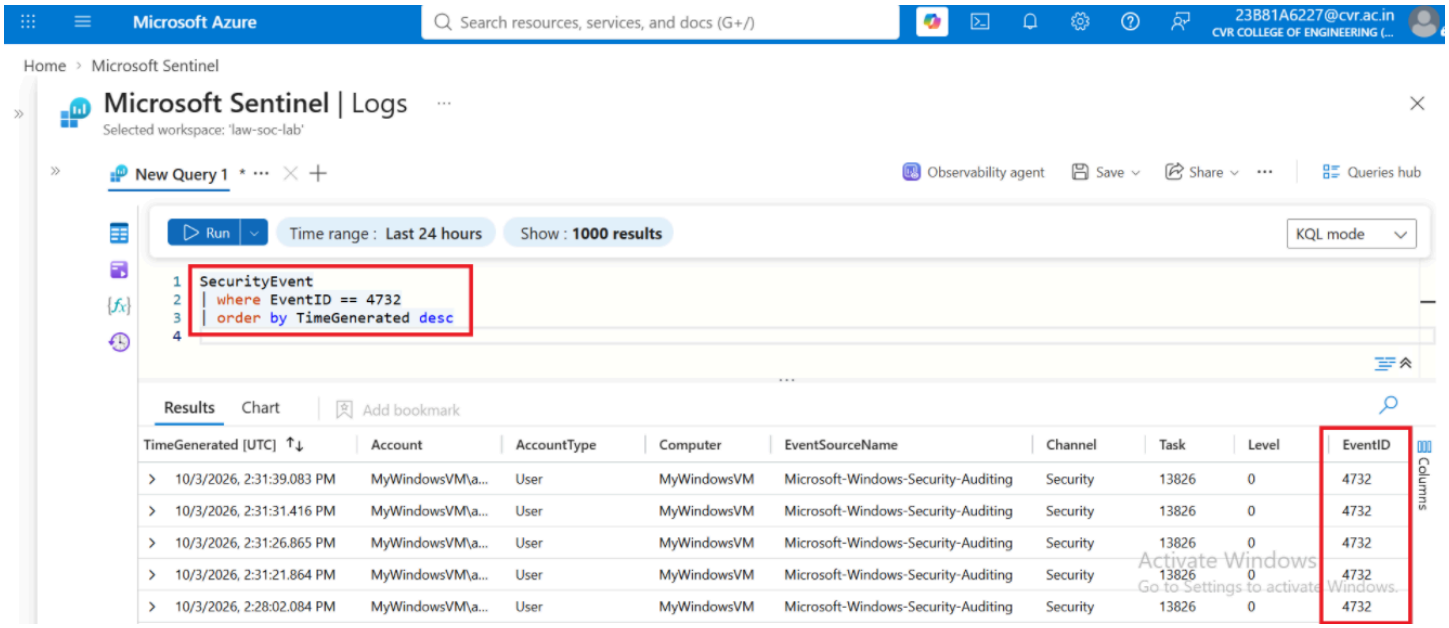
```
SecurityEvent
| take 10
```



    This displays 10 Windows Security events collected from the VM

22. To verify the **Event ID 4732** generated earlier, run:

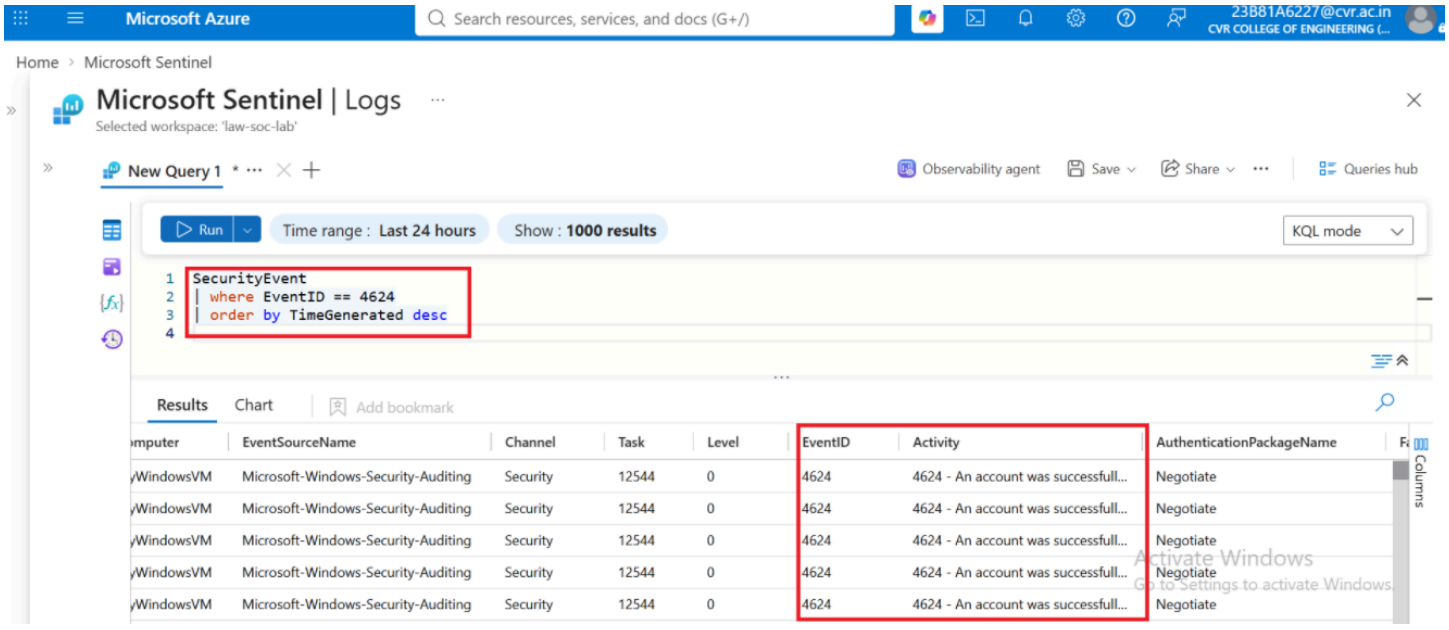
```
SecurityEvent
| where EventID == 4732
| order by TimeGenerated desc
```



This displays the most recent events where a user was added to a local security-enabled group.

23. To verify the successful login made using `soc_user01`, run:

```
SecurityEvent
| where EventID == 4624
| order by TimeGenerated desc
```



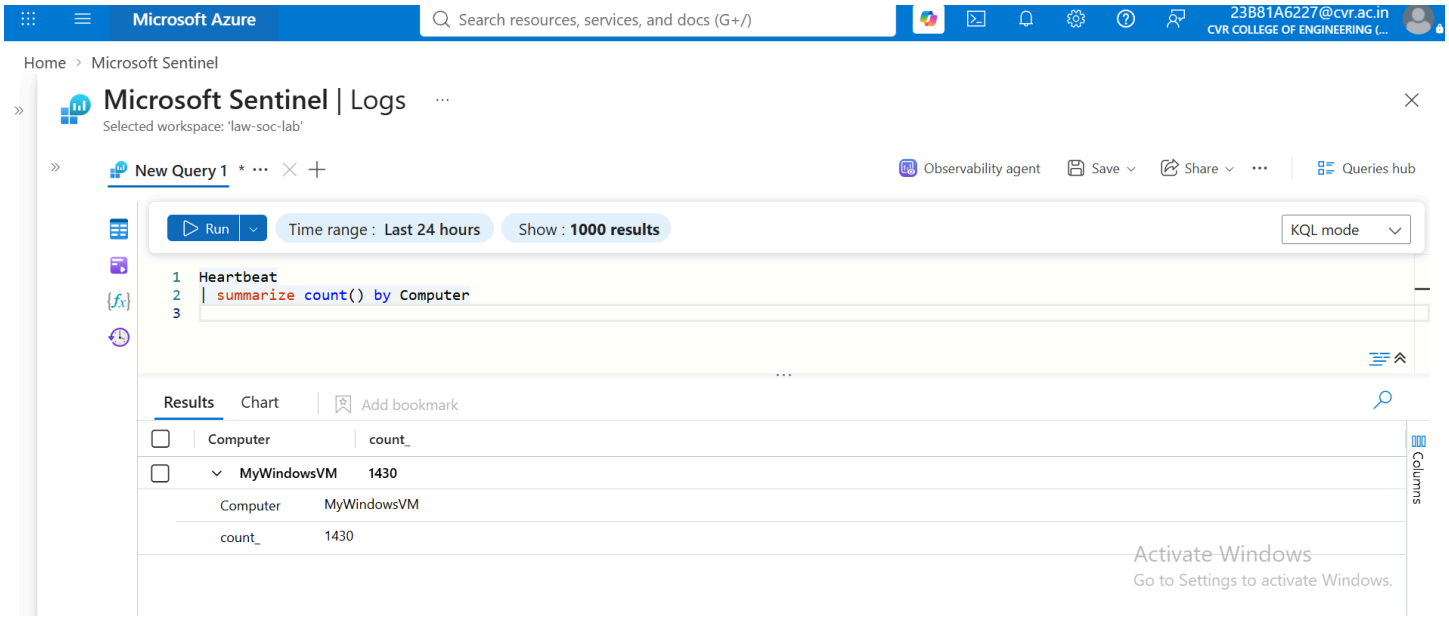
Event ID 4624 represents a successful logon.

VII. Prerequisites Before Simulating Attacks

1. **Check Heartbeat Status** (Verify Sentinel before attacks)

To check whether the VM is sending heartbeat information to Sentinel, run:

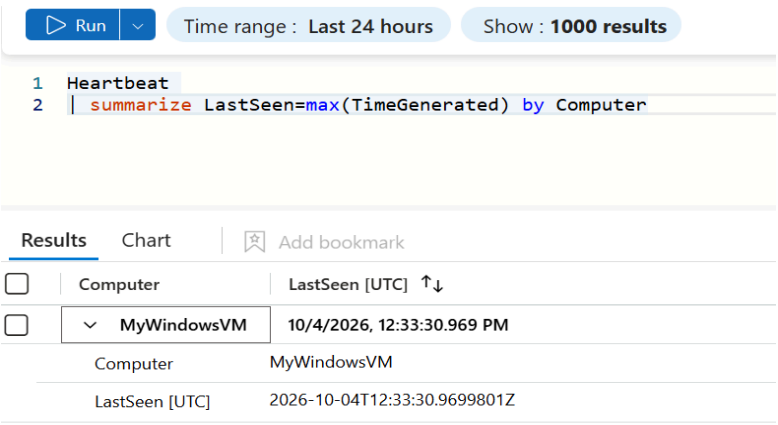
```
Heartbeat
| summarize count() by Computer
```



This means:

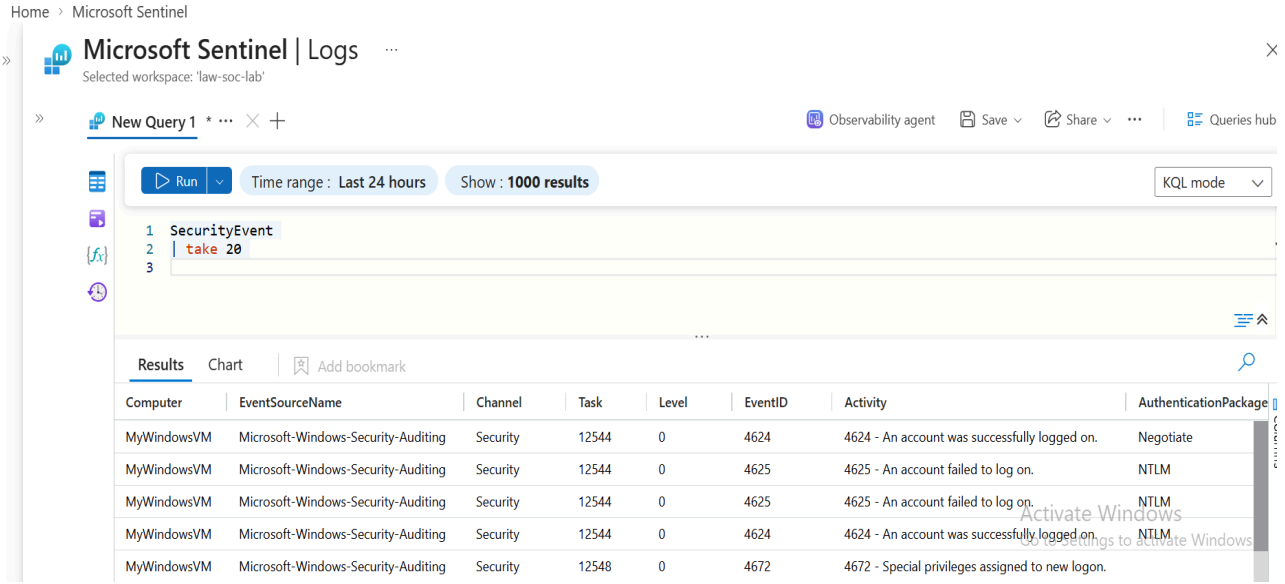
- **Computer:** `MyWindowsVM` → this is your Windows VM.
- **count_:** `1430` → Sentinel has received **1,430 heartbeat records** from `MyWindowsVM` within the selected **Last 24 hours** time range.

```
Heartbeat
| summarize LastSeen=max(TimeGenerated) by Computer
```



2. Check SecurityEvent

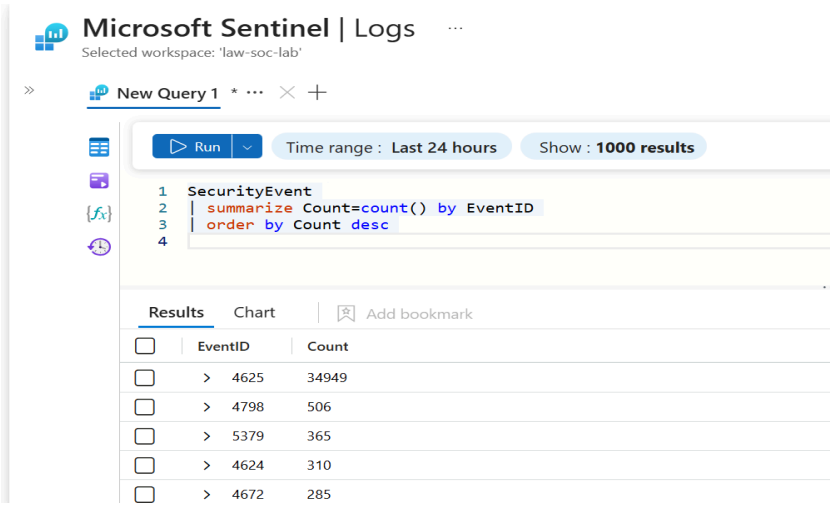
```
SecurityEvent
| take 20
```



If you get records, your Windows Security Event ingestion is working.

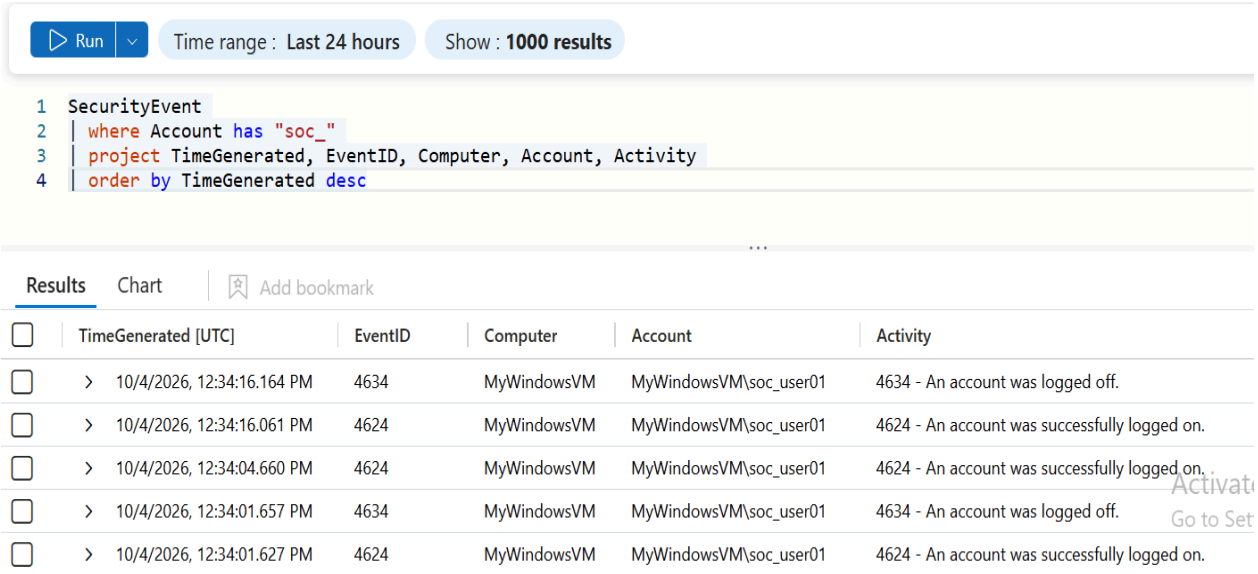
3. **Check the event IDs:**

```
SecurityEvent
| summarize Count=count() by EventID
| order by Count desc
```



4. **Check the current user activity:**

```
SecurityEvent
| where Account has "soc_"
| project TimeGenerated, EventID, Computer, Account, Activity
| order by TimeGenerated desc
```



VIII. Attack Simulations

1. Failed Login / Brute-Force Simulation

- Open **Remote Desktop / Windows App** and connect to `MyWindowsVM`.
- At the Windows login screen, select **Use another account**.
- Enter:
  - Username: soc_user01
- Enter an intentionally incorrect password:
  - WrongPassword@123
- Click **Sign in**.
- Repeat the failed login approximately **5 times**.
- The expected event is **4625 — An account failed to log on**. MS Sentinel
- Open **Microsoft Sentinel → law-soc-lab → Logs**.
- Run:

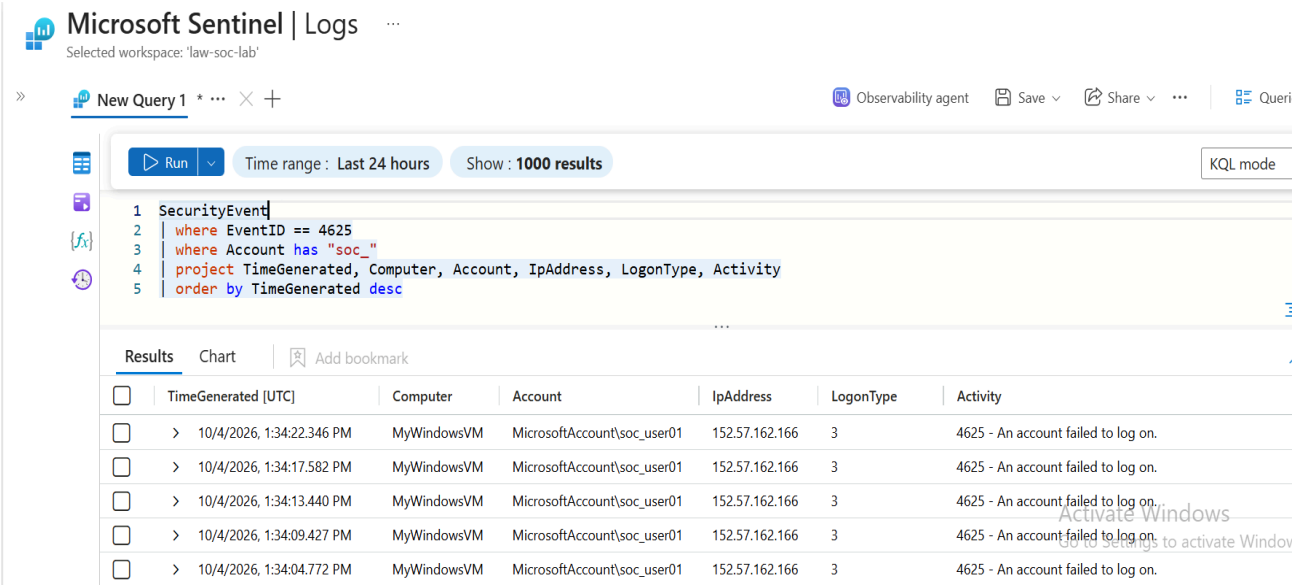
```
SecurityEvent

| where EventID == 4625

| where Account has "soc_"

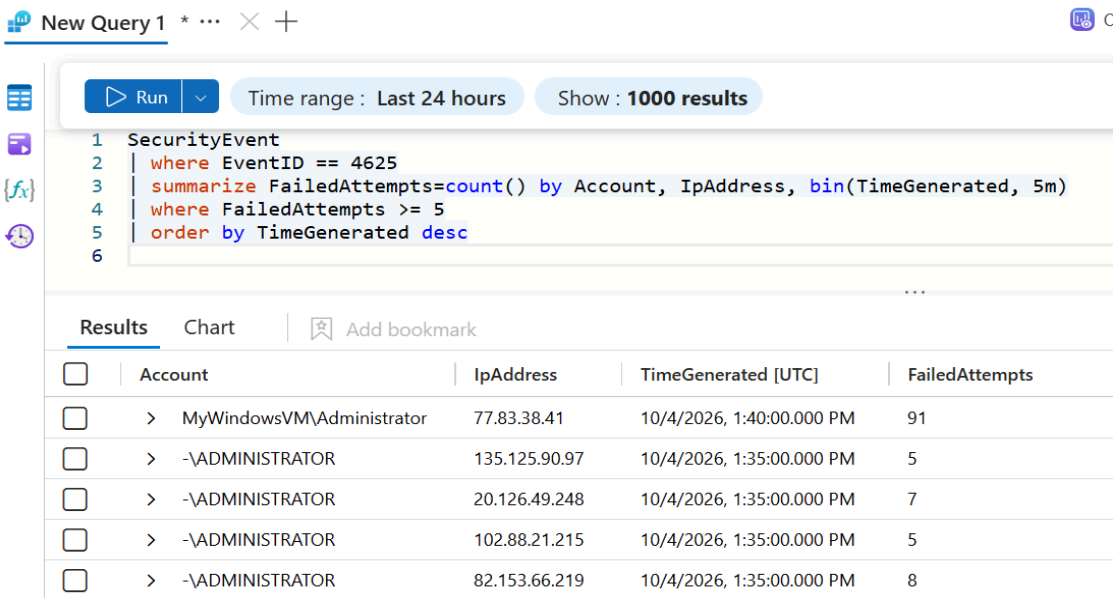
| project TimeGenerated, Computer, Account, IpAddress, LogonType, Activity

| order by TimeGenerated desc
```



- To detect 5 or more failed attempts within 5 minutes, run:

```
SecurityEvent
| where EventID == 4625
| summarize FailedAttempts=count() by Account, IpAddress, bin(TimeGenerated, 5m)
| where FailedAttempts >= 5
| order by TimeGenerated desc
```



This identifies a **potential brute-force activity** when the same account has 5 or more failures within a 5-minute window. This is a detection hypothesis requiring investigation. It's **not an attack.**

2. **Failed Login → Successful Login**

- After generating the failed login attempts, connect to `MyWindowsVM` again.

- Log in using:
  - Username: soc_user01
  - Enter the **correct password** for `soc_user01`.
- Open **Microsoft Sentinel → law-soc-lab → Logs**.
- Run:

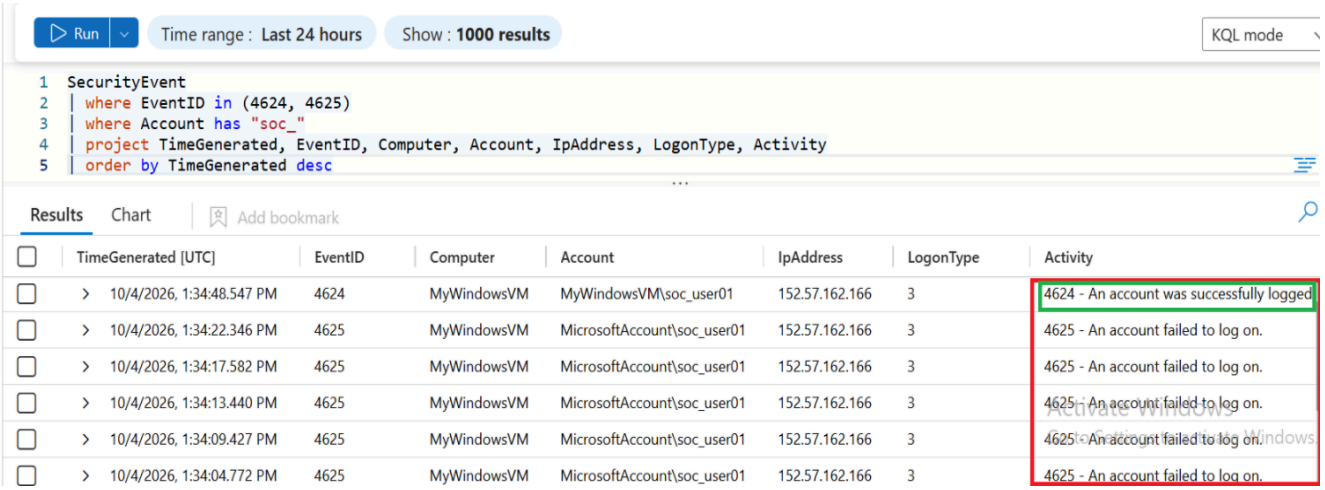
  SecurityEvent

  | where EventID in (4624, 4625)

  | where Account has "soc_"

  | project TimeGenerated, EventID, Computer, Account, IpAddress, LogonType, Activity

  | order by TimeGenerated desc



- Check the timeline of events. You see a sequence similar to:

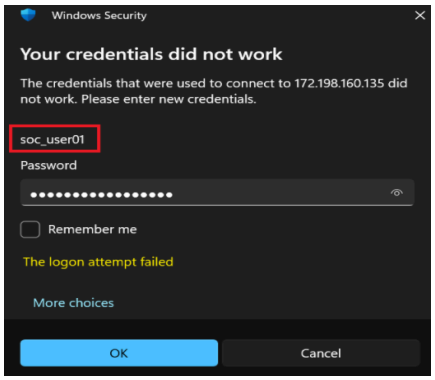
4625	Failed
4625	Failed
4625	Failed
4625	Failed
4624	Successful

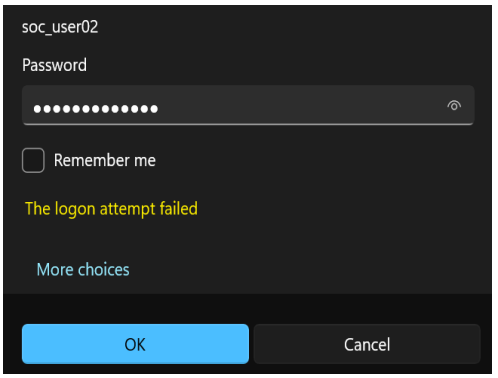
- This lets you investigate a **failed-login sequence followed by a successful login**.

3. Password Spray Simulation

- Open the Windows login interface.
- Attempt to log in using:
  - soc_user01
  - soc_user02
  - soc_user03
  - soc_user04
- Use the **same deliberately incorrect password** for each account as **WrongPassword@123**
- The concept being demonstrated is:

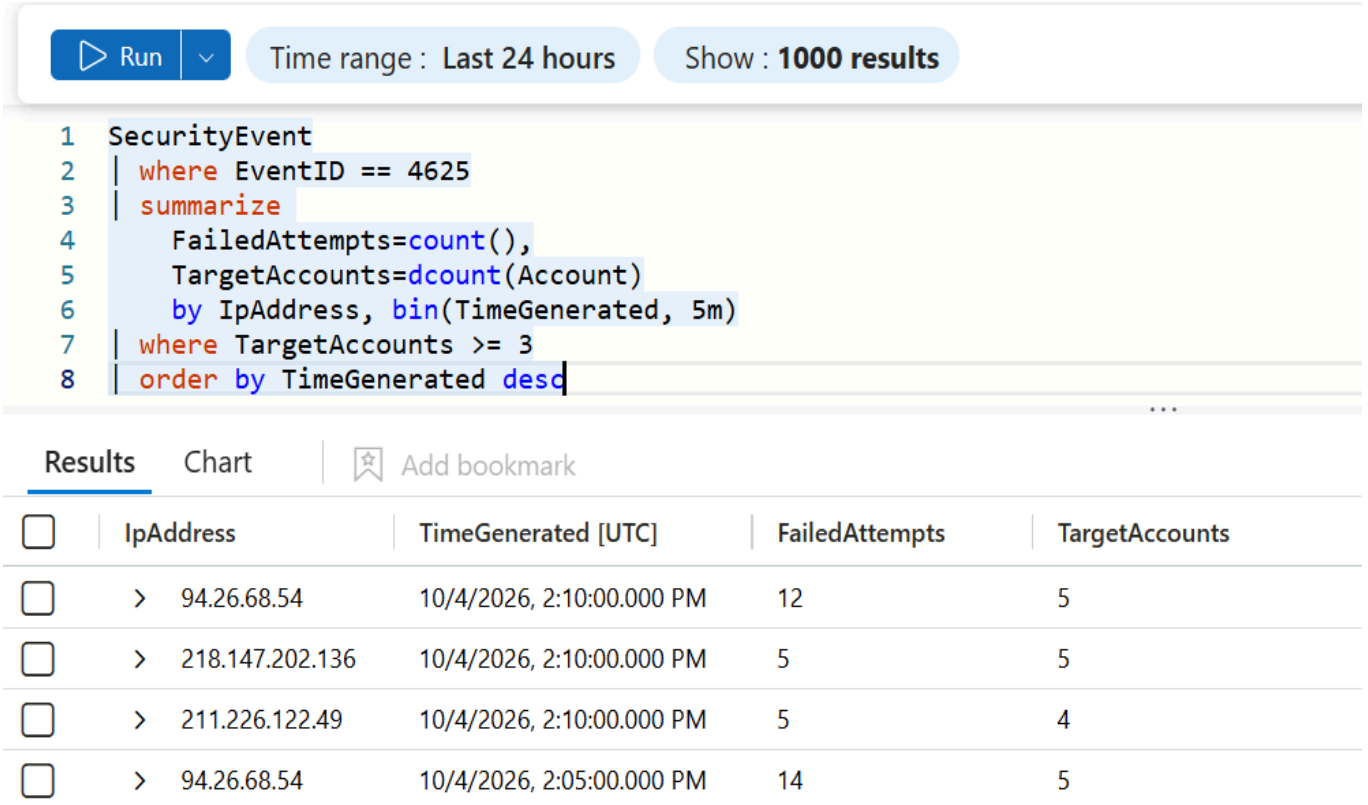
- **Brute Force:** One account → Many passwords
- **Password Spray:** One password → Many accounts





(do the same with other users too)

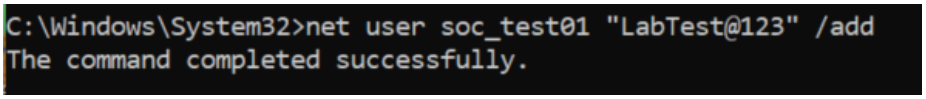
- Then open **Microsoft Sentinel → law-soc-lab → Logs**.
- Run:



- Check the `TargetAccounts` value.
- `dcount(Account)` counts the number of **distinct accounts** targeted.

4. Unauthorized Account Creation

- Open **Command Prompt** as Administrator.
- Create a temporary test account: **net user soc_test01 "LabTest@123" /add**



- This should generate **Event ID 4720 — User account created**.
- Open **Microsoft Sentinel → law-soc-lab → Logs**.

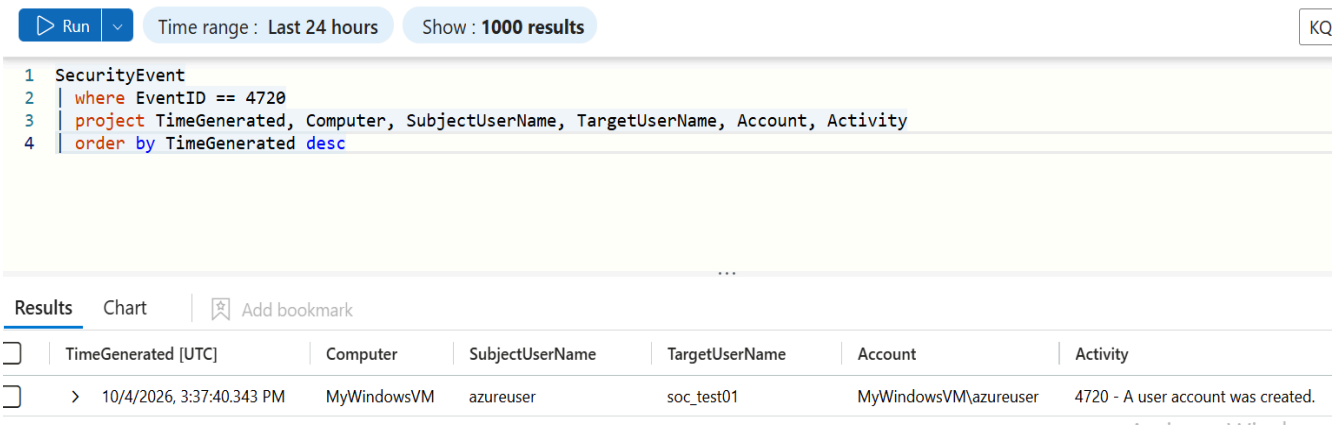
- Run:

  SecurityEvent

  | where EventID == 4720

  | project TimeGenerated, Computer, SubjectUserName, TargetUserName, Account, Activity

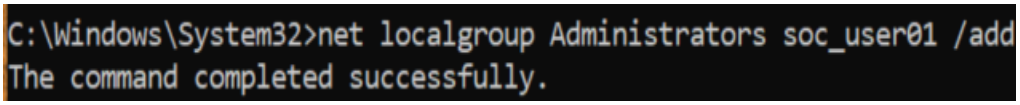
  | order by TimeGenerated desc



  - Then SOC Investigates: **Who created `soc_test01`?**

5. Privilege Escalation

- Open **Command Prompt** as Administrator.
- Add the standard user `soc_user01` to the local Administrators group: **net localgroup Administrators soc_user01 /add**



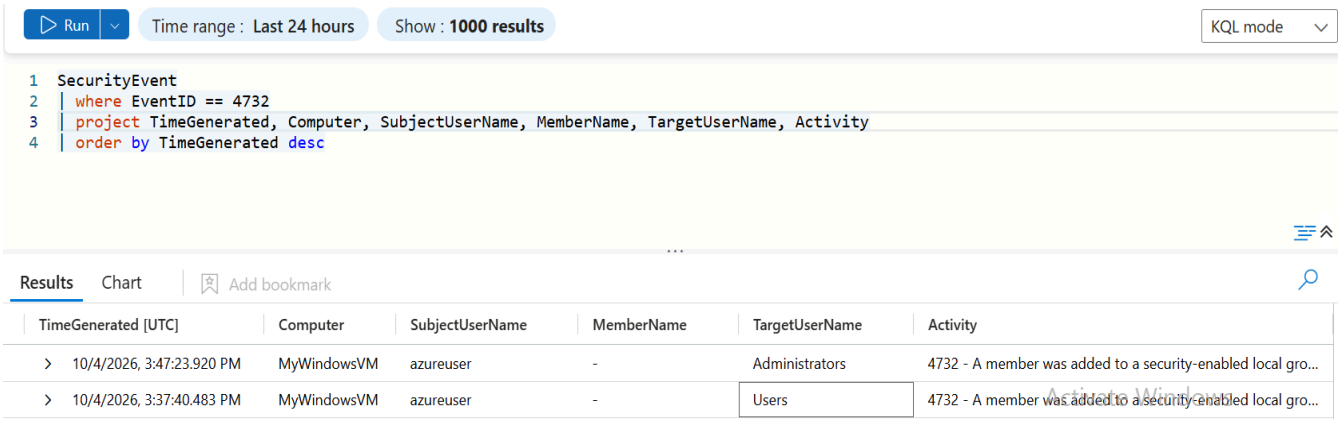
- This should generate **Event ID 4732 — Member added to a security-enabled local group**. MS Sentinel
- Open **Microsoft Sentinel → law-soc-lab → Logs**.
- Run:

  SecurityEvent

  | where EventID == 4732

  | project TimeGenerated, Computer, SubjectUserName, MemberName, TargetUserName, Activity

  | order by TimeGenerated desc

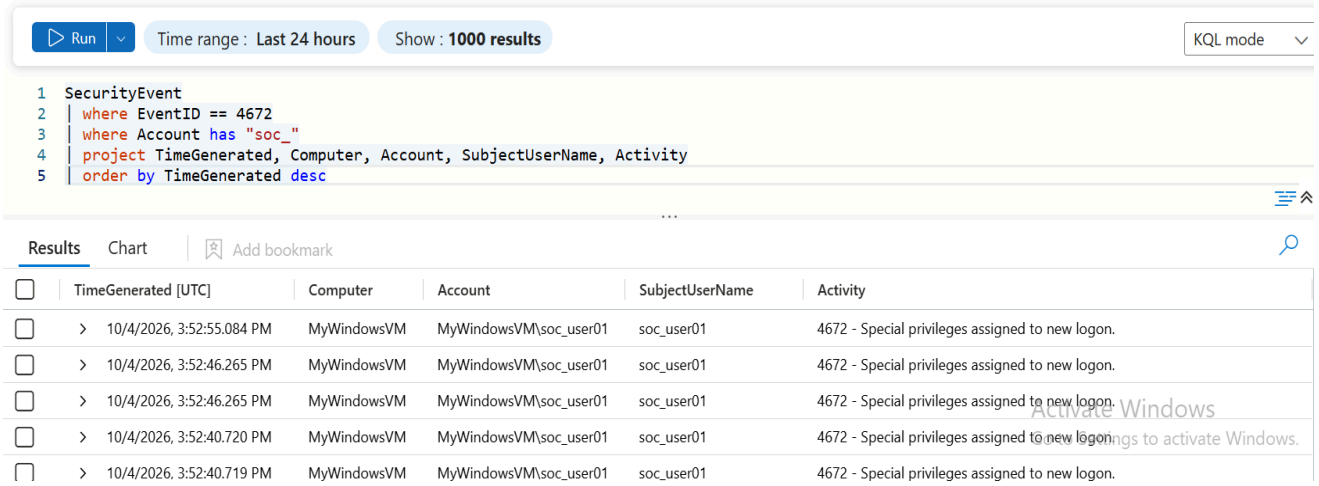


- Investigate: **Why was a standard account added to the local Administrators group?**

6. **Privileged Logon**

- Log out of the Windows VM.
- Log in using:
  - Username: soc_user01
- Since `soc_user01` was added to the Administrators group in the previous attack, this is now a privileged account.
- Open **Microsoft Sentinel → law-soc-lab → Logs**.
- Run:

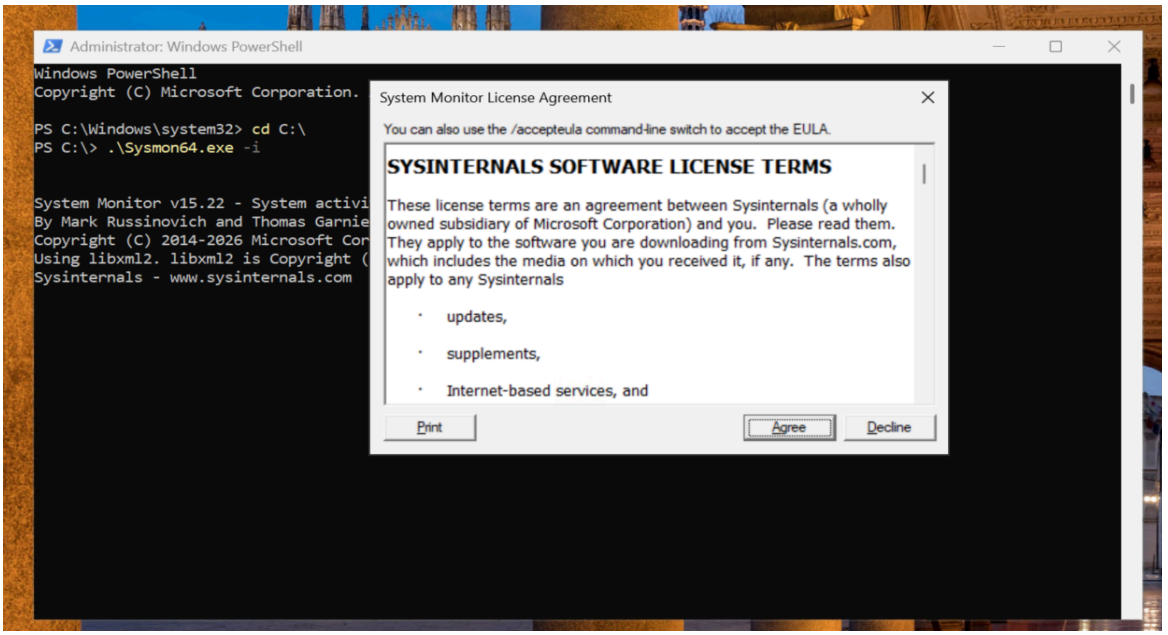
```
SecurityEvent
| where EventID == 4672
| where Account has "soc_"
| project TimeGenerated, Computer, Account, SubjectUserName, Activity
| order by TimeGenerated desc
```



Event **4672** indicates that special privileges were assigned to a new logon.

7. Process Creation

- Download Sysmon, extract it into **C:** folder
- Open **PowerShell** as Administrator.
- Type the command:
    - **cd C:/**
    - **.\Sysmon64.exe -i**
    - **Click Agree**



- Check whether Sysmon is running: **Get-Service Sysmon64**

```
PS C:\> Get-Service Sysmon64

Status   Name               DisplayName
------   ----               -----------
Running  Sysmon64           Sysmon64
```

- Check recent Sysmon events: **Get-WinEvent -LogName "Microsoft-Windows-Sysmon/Operational" -MaxEvents 10**

```
PS C:\> Get-WinEvent -LogName "Microsoft-Windows-Sysmon/Operational" -MaxEvents 10


   ProviderName: Microsoft-Windows-Sysmon

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
10/4/2026 4:46:20 PM             5 Information      Process terminated:...
10/4/2026 4:46:03 PM             5 Information      Process terminated:...
10/4/2026 4:46:03 PM             5 Information      Process terminated:...
10/4/2026 4:44:49 PM             1 Information      Process Create:...
10/4/2026 4:44:36 PM             5 Information      Process terminated:...
10/4/2026 4:44:05 PM             5 Information      Process terminated:...
10/4/2026 4:43:58 PM             5 Information      Process terminated:...
10/4/2026 4:43:55 PM             5 Information      Process terminated:...
10/4/2026 4:43:47 PM             5 Information      Process terminated:...
10/4/2026 4:43:16 PM             1 Information      Process Create:...
```

- Generate harmless process activity:
  - **notepad.exe**
  - **whoami**
  - **ipconfig**

```
PS C:\> notepad.exe
PS C:\> whoami
mywindowsvm\azureuser
PS C:\> ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : blxl4lswcr0ebm0p1hf35ttajf.hydx.internal.cloudapp.net
   Link-local IPv6 Address . . . . . : fe80::4911:191c:d912:32f5%4
   IPv4 Address. . . . . . . . . . . : 172.16.0.4
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 172.16.0.1
PS C:\> _
```

- Then:
  - **Get-Process**

```
PS C:\> Get-Process

Handles  NPM(K)    PM(K)      WS(K)     CPU(s)      Id  SI ProcessName
-------  ------    -----      -----     ------      --  -- -----------
    136       8     1740          4       2.13    4020   0 AggregatorHost
    190      14     3108       2148      17.63    5116   0 AMAExtHealthMonitor
    352      19    11900       1000      10.22   12296   3 AppActions
    830      27    26152       2412       4.23   12372   3 backgroundTaskHost
     76       8     1052          0       0.03    3552   0 conhost
     80       8     1228          4       0.58    4176   0 conhost
     76       8     1136          4       0.19    4216   0 conhost
    283      16     3032      12168       0.97    6316   3 conhost
    486      25     8784          4       2.11    6672   2 CrossDeviceResume
    501      24     8840         68       5.98   10160   3 CrossDeviceResume
    597      22     2440        184       7.59     648   0 csrss
    156      12     2012        264       2.72     724   1 csrss
    549      22     2652       9528      32.66    4552   3 csrss
    389      18     2400        572       6.36    5272   2 csrss
    549      19     5732       6584       7.41    1144   3 ctfmon
    497      18     4896          0       0.84    7780   2 ctfmon
```

- Search specifically for Sysmon Process Creation events:

  Get-WinEvent -FilterHashtable @{

  LogName='Microsoft-Windows-Sysmon/Operational'

  Id=1

  } -MaxEvents 20

```
PS C:\> Get-WinEvent -FilterHashTable @{
>> LogName='Microsoft-Windows-Sysmon/Operational'
>> Id=1
>> } -MaxEvents 20


   ProviderName: Microsoft-Windows-Sysmon

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
10/4/2026 4:54:03 PM             1 Information      Process Create:...
10/4/2026 4:54:01 PM             1 Information      Process Create:...
10/4/2026 4:53:35 PM             1 Information      Process Create:...
10/4/2026 4:52:59 PM             1 Information      Process Create:...
10/4/2026 4:52:52 PM             1 Information      Process Create:...
10/4/2026 4:49:45 PM             1 Information      Process Create:...
10/4/2026 4:49:40 PM             1 Information      Process Create:...
10/4/2026 4:49:15 PM             1 Information      Process Create:...
10/4/2026 4:49:08 PM             1 Information      Process Create:...
10/4/2026 4:49:07 PM             1 Information      Process Create:...
10/4/2026 4:49:03 PM             1 Information      Process Create:...
10/4/2026 4:49:03 PM             1 Information      Process Create:...
10/4/2026 4:44:49 PM             1 Information      Process Create:...
10/4/2026 4:43:16 PM             1 Information      Process Create:...
10/4/2026 4:43:15 PM             1 Information      Process Create:...
10/4/2026 4:43:14 PM             1 Information      Process Create:...
10/4/2026 4:43:10 PM             1 Information      Process Create:...
10/4/2026 4:43:10 PM             1 Information      Process Create:...
```

- Sysmon **Event ID 1** records process creation.

8. PowerShell Investigation

- Open **PowerShell**.
- Run:
  - Get-Process

```
PS C:\> Get-Process

Handles  NPM(K)    PM(K)      WS(K)     CPU(s)      Id  SI ProcessName
-------  ------    -----      -----     ------      --  -- -----------
    136       8     1740         24       2.17    4020   0 AggregatorHost
    200      15     3924       1364      17.78    5116   0 AMAExtHealthMonitor
    352      19    11904       4904      10.30   12296   3 AppActions
    228      12     3308       2020       0.27    9600   2 backgroundTaskHost
    830      27    26148        304       4.27   12372   3 backgroundTaskHost
     76       8     1052          0       0.03    3552   0 conhost
     80       8     1228          4       0.58    4176   0 conhost
     76       8     1136          8       0.19    4216   0 conhost
    283      16    10108      11412       1.95    6316   3 conhost
    486      25     8784         28       2.13    6672   2 CrossDeviceResume
    501      24     8844        528       6.03   10160   3 CrossDeviceResume
    603      22     2440        632       7.67     648   0 csrss
    156      12     2012         32       2.72     724   1 csrss
    548      22     2648        228      32.77    4552   3 csrss
    393      18     2404        140       6.36    5272   2 csrss
    549      19     5792       1536       7.47    1144   3 ctfmon
    497      18     4896          0       0.84    7780   2 ctfmon
```

  - Get-Service

```
PS C:\> Get-Service

Status   Name               DisplayName
------   ----               -----------
Stopped  AarSvc_218b20      Agent Activation Runtime_218b20
Stopped  AarSvc_b6835       Agent Activation Runtime_b6835
Stopped  ADPSvc             Aggregated Data Platform Service
Stopped  ALG                Application Layer Gateway Service
Running  AppIDSvc           Application Identity
Running  Appinfo            Application Information
Stopped  AppMgmt            Application Management
Stopped  AppReadiness       App Readiness
Stopped  AppVClient         Microsoft App-V Client
Running  AppXSvc            AppX Deployment Service (AppXSVC)
Stopped  ApxSvc             Windows Virtual Audio Device Proxy ...
```

- Investigate the corresponding process telemetry in Sysmon/Sentinel.

9. File Creation

- Open **PowerShell**.
- Create a temporary directory:
  - **New-Item C:\Temp -ItemType Directory -Force**

```
PS C:\Windows\system32> New-Item C:\Temp -ItemType Directory -Force


    Directory: C:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         10/4/2026   5:20 PM                Temp
```

- Create a test file:
  - "Sentinel SOC Lab Test" | Out-File C:\Temp\soc-test.txt

```
PS C:\Windows\system32> "Sentinel SOC Lab Test" | Out-File C:\Temp\soc-test.txt
```

- This generates file-creation activity. Sysmon **Event ID 11** records file creation/overwrite activity. MS Sentinel
- Verify locally using:

Get-WinEvent -FilterHashtable @{

LogName='Microsoft-Windows-Sysmon/Operational'

Id=11

} -MaxEvents 20

```
PS C:\Windows\system32> Get-WinEvent -FilterHashtable @{
>> LogName='Microsoft-Windows-Sysmon/Operational'
>> Id=11
>> } -MaxEvents 20


   ProviderName: Microsoft-Windows-Sysmon

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
10/5/2026 4:03:23 AM            11 Information      File created:...
10/5/2026 2:40:09 AM            11 Information      File created:...
10/5/2026 2:39:48 AM            11 Information      File created:...
10/5/2026 1:24:24 AM            11 Information      File created:...
10/5/2026 1:24:23 AM            11 Information      File created:...
```

- Investigate: Which process created the file, under which account, and where?

10. DNS Investigation

- Open **PowerShell**.
- Run:
  - **nslookup example.com**

```
PS C:\Windows\system32> nslookup example.com
Server:  UnKnown
Address:  168.63.129.16

Non-authoritative answer:
Name:    example.com
Addresses:  2606:4700:10::ac42:93f3
          2606:4700:10::6814:179a
          172.66.147.243
          104.20.23.154
```

- This generates DNS query activity.
- Sysmon **Event ID 22** records DNS query activity. MS Sentinel
- Then look for the DNS event:

Get-WinEvent -FilterHashtable @{

LogName='Microsoft-Windows-Sysmon/Operational'

Id=22

} -MaxEvents 20

```
PS C:\Windows\system32> Get-WinEvent -FilterHashtable @{
>> LogName='Microsoft-Windows-Sysmon/Operational'
>> Id=22
>> } -MaxEvents 20


   ProviderName: Microsoft-Windows-Sysmon

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
10/5/2026 4:21:18 AM            22 Information      Dns query:...
10/5/2026 3:53:21 AM            22 Information      Dns query:...
10/5/2026 3:21:05 AM            22 Information      Dns query:...
10/5/2026 2:20:54 AM            22 Information      Dns query:...
10/5/2026 1:20:42 AM            22 Information      Dns query:...
10/5/2026 12:20:30 AM           22 Information      Dns query:...
10/4/2026 11:20:18 PM           22 Information      Dns query:...
10/4/2026 10:20:07 PM           22 Information      Dns query:...
10/4/2026 9:39:49 PM            22 Information      Dns query:...
10/4/2026 9:29:47 PM            22 Information      Dns query:...
10/4/2026 9:28:02 PM            22 Information      Dns query:...
10/4/2026 9:19:55 PM            22 Information      Dns query:...
10/4/2026 9:14:45 PM            22 Information      Dns query:...
10/4/2026 8:52:29 PM            22 Information      Dns query:...
10/4/2026 8:19:43 PM            22 Information      Dns query:...
10/4/2026 7:49:12 PM            22 Information      Dns query:...
10/4/2026 7:29:22 PM            22 Information      Dns query:...
10/4/2026 7:19:31 PM            22 Information      Dns query:...
10/4/2026 7:14:45 PM            22 Information      Dns query:...
10/4/2026 7:14:45 PM            22 Information      Dns query:...
```

- **Result:** `nslookup` successfully resolved `example.com`, and Sysmon Event ID 22 confirmed that the DNS query was recorded. This demonstrates basic DNS monitoring and investigation using Sysmon.

11. Network Connection

- Open **PowerShell**.
- Run the following benign connection test:
  - Test-NetConnection example.com -Port 443

```
PS C:\Windows\system32> Test-NetConnection example.com -Port 443


ComputerName     : example.com
RemoteAddress    : 172.66.147.243
RemotePort       : 443
InterfaceAlias   : Ethernet
SourceAddress    : 172.16.0.4
TcpTestSucceeded : True
```

- This tests connectivity to `example.com` on TCP port `443`.
- Verify if the network activity was recorded:

Get-WinEvent -FilterHashtable @{

LogName='Microsoft-Windows-Sysmon/Operational'

Id=3

} -MaxEvents 10

```
PS C:\Windows\system32> Get-WinEvent -FilterHashtable @{
>> LogName='Microsoft-Windows-Sysmon/Operational'
>> Id=3
>> } -MaxEvents 10


   ProviderName: Microsoft-Windows-Sysmon

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
10/5/2026 4:35:33 AM             3 Information      Network connection detected:...
10/5/2026 4:35:31 AM             3 Information      Network connection detected:...
10/5/2026 4:35:31 AM             3 Information      Network connection detected:...
10/5/2026 4:35:30 AM             3 Information      Network connection detected:...
10/5/2026 4:35:30 AM             3 Information      Network connection detected:...
10/5/2026 4:35:29 AM             3 Information      Network connection detected:...
10/5/2026 4:35:29 AM             3 Information      Network connection detected:...
10/5/2026 4:35:28 AM             3 Information      Network connection detected:...
10/5/2026 4:35:28 AM             3 Information      Network connection detected:...
10/5/2026 4:35:27 AM             3 Information      Network connection detected:...
```

- Sysmon **Event ID 3** can record network connections when it is enabled in the Sysmon configuration.

12. Registry Modification

- Open **Command Prompt**.
- Create a dedicated test registry key:
  - **reg add HKCU\Software\SOC-Lab /v TestValue /t REG_SZ /d "SentinelDemo" /f**

```
C:\Windows\System32>reg add HKCU\Software\SOC-Lab /v TestValue /t REG_SZ /d "SentinelDemo" /f
The operation completed successfully.
```

- **Meaning of the above command:**

`reg`	Windows command used to manage the Registry
`add`	Add a new registry key/value
`HKCU`	**HKEY_CURRENT_USER** — registry settings for the current user
`\Software\SOC-Lab`	Registry key where we're adding the value
`/v TestValue`	Creates a value named **TestValue**
`/t REG_SZ`	Specifies that the value is a normal text/string
`/d "SentinelDemo"`	The actual data stored in the value
`/f`	Forces the operation without asking for confirmation

- This creates/modifies a registry value under the current user's registry hive.
- Sysmon Event IDs **12–14** cover registry activity.
- Verify:

```
PS C:\Windows\system32> Get-WinEvent -FilterHashtable @{
>> LogName='Microsoft-Windows-Sysmon/Operational'
>> Id=12,13,14
>> } -MaxEvents 20


   ProviderName: Microsoft-Windows-Sysmon

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
10/5/2026 4:59:02 AM            13 Information      Registry value set:...
10/5/2026 4:59:02 AM            13 Information      Registry value set:...
10/5/2026 3:59:44 AM            13 Information      Registry value set:...
10/5/2026 3:59:41 AM            13 Information      Registry value set:...
10/5/2026 3:59:41 AM            13 Information      Registry value set:...
10/5/2026 3:58:53 AM            13 Information      Registry value set:...
10/5/2026 3:58:51 AM            13 Information      Registry value set:...
10/5/2026 3:58:04 AM            12 Information      Registry object added or deleted:...
10/5/2026 3:58:04 AM            12 Information      Registry object added or deleted:...
10/5/2026 3:58:03 AM            12 Information      Registry object added or deleted:...
10/5/2026 3:56:32 AM            13 Information      Registry value set:...
10/5/2026 3:56:30 AM            13 Information      Registry value set:...
10/5/2026 3:56:30 AM            13 Information      Registry value set:...
10/5/2026 3:55:38 AM            12 Information      Registry object added or deleted:...
10/5/2026 3:55:38 AM            12 Information      Registry object added or deleted:...
10/5/2026 3:55:36 AM            12 Information      Registry object added or deleted:...
10/5/2026 3:53:19 AM            13 Information      Registry value set:...
10/5/2026 3:53:18 AM            13 Information      Registry value set:...
10/5/2026 3:53:18 AM            13 Information      Registry value set:...
10/5/2026 3:51:48 AM            12 Information      Registry object added or deleted:...
```

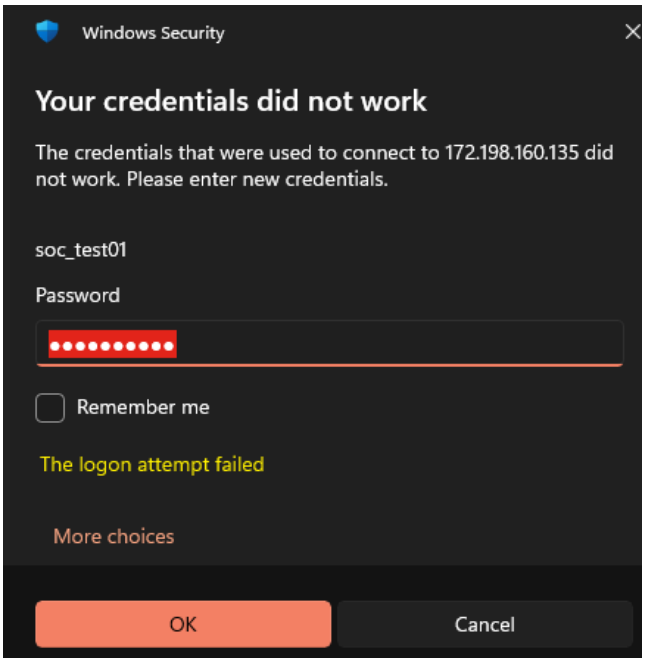
13. Account Lockout

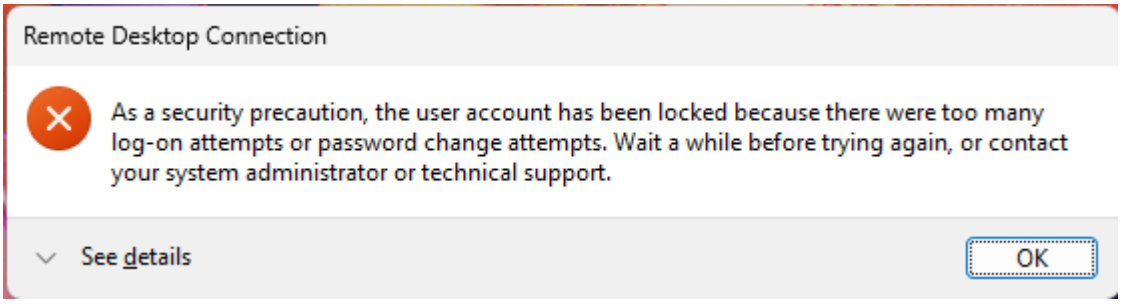
- Use a **lab account only**.

  Username: soc_test01

  Password: enter wrong password

- Generate enough incorrect authentication attempts to trigger the configured Windows account-lockout policy. (10-11 wrong password attempts)





- After the account is locked, open **Microsoft Sentinel → law-soc-lab → Logs**.
- Run:

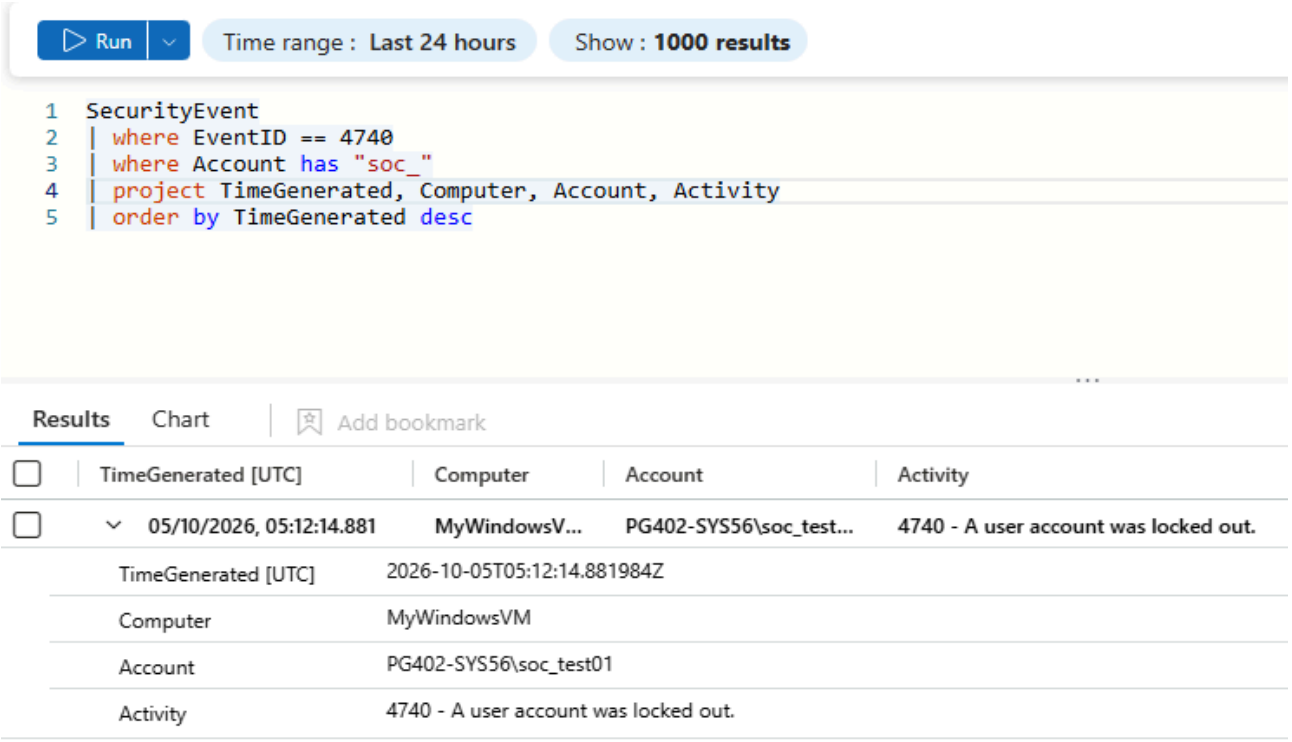
  SecurityEvent

  | where EventID == 4740

  | where Account has "soc_"

  | project TimeGenerated, Computer, Account, Activity

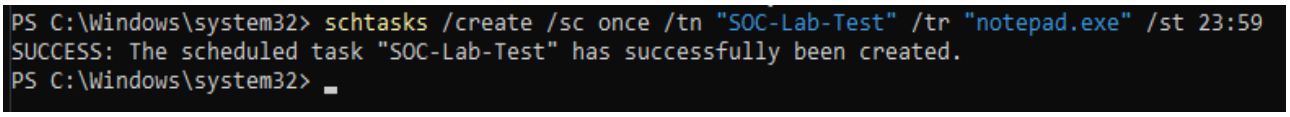
  | order by TimeGenerated desc



- Event **4740** represents an account lockout.

14. Scheduled Task

- Open **Poweshell** as Administrator.
- Create a harmless scheduled task:
  - **schtasks /create /sc once /tn "SOC-Lab-Test" /tr "notepad.exe" /st 23:59**



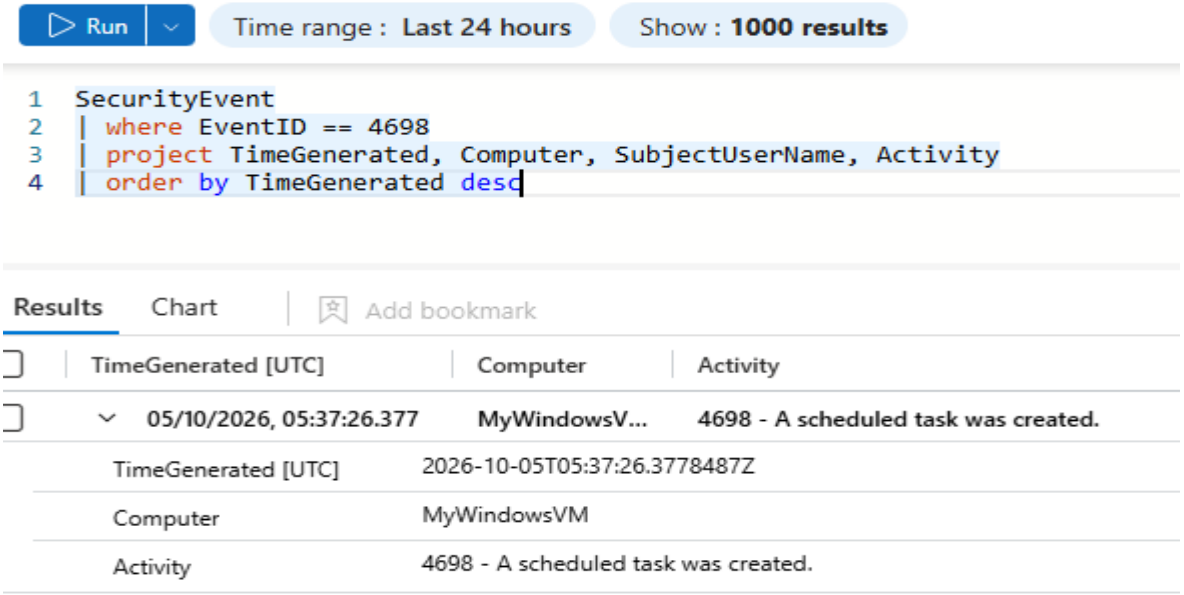
- This creates a scheduled task named `SOC-Lab-Test` that is configured to run `notepad.exe`.
- Open **Microsoft Sentinel → law-soc-lab → Logs**.
- Search for:

  SecurityEvent

  | where EventID == 4698

  | project TimeGenerated, Computer, SubjectUserName, Activity

  | order by TimeGenerated desc



- Event **4698** represents scheduled-task creation. MS Sentinel
- After the demonstration, delete the test task:
    - **schtasks /delete /tn "SOC-Lab-Test" /f**

```
PS C:\Windows\system32> schtasks /delete /tn "SOC-Lab-Test" /f
SUCCESS: The scheduled task "SOC-Lab-Test" was successfully deleted.
```

15. Security Log Clearing

- **Perform this only on your isolated lab VM.**
- The objective is to demonstrate that an attempt to clear the Windows Security log can itself generate a security event.
- Open **PowerShell as Administrator**.
- Run:
    - **wevtutil cl Security**

```
PS C:\Windows\system32> wevtutil cl Security
PS C:\Windows\system32>
```

- Check locally for Event 1102, using the command:

Get-WinEvent -FilterHashtable @{

LogName='Security'

Id=1102

} -MaxEvents 5 | Format-List TimeCreated, Id, Message

```
PS C:\Windows\system32> Get-WinEvent -FilterHashtable @{
>> LogName='Security'
>> Id=1102
>> } -MaxEvents 10 | Format-List TimeCreated, Id, Message


TimeCreated : 10/5/2026 5:49:20 AM
Id          : 1102
Message     : The audit log was cleared.
              Subject:
                Security ID:    S-1-5-21-2322504839-385044276-2162829752-500
                Account Name:   azureuser
                Domain Name:    MyWindowsVM
                Logon ID:       0x4659B19
```

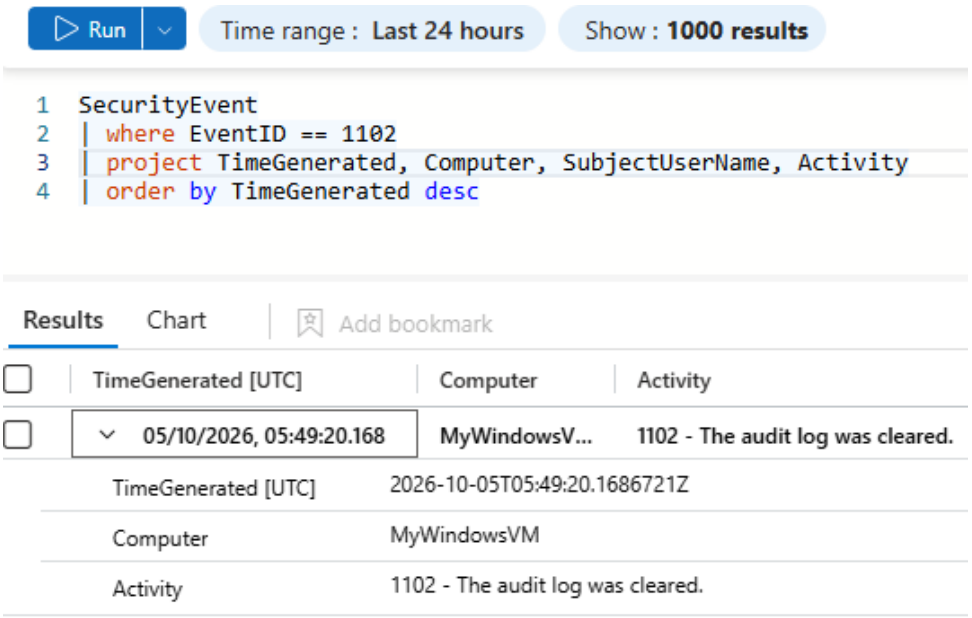
- After performing the lab demonstration, open **Microsoft Sentinel → law-soc-lab → Logs**.
- Search for:

  SecurityEvent

  | where EventID == 1102

  | project TimeGenerated, Computer, SubjectUserName, Activity

  | order by TimeGenerated desc



- Event **1102** indicates that the Security audit log was cleared.

IX. Master SOC Investigation

Scenario: Suspicious Administrator Activity

An administrator account on the Windows VM may have been compromised. Determine what happened by correlating the Windows Security events generated during the lab.

1. Open Microsoft Sentinel

- Open the Azure Portal.
- Search for Microsoft Sentinel.
- Open law-soc-lab.
- Click Logs.

2. Run the Master Investigation Query

- Enter the following KQL query:

  SecurityEvent

  | where Account has "soc_"

  | where EventID in (4624, 4625, 4648, 4672, 4688, 4697, 4720, 4728, 4732, 4740, 1102)

  | project TimeGenerated, EventID, Computer, Account, SubjectUserName, TargetUserName, Activity

  | order by TimeGenerated asc

```
SecurityEvent
| where Account has "soc_"
| where EventID in ( 4624, 4625, 4648, 4672, 4688, 4697, 4720, 4728, 4732, 4740, 1102)
| project TimeGenerated, EventID, Computer, Account, SubjectUserName, TargetUserName, Activity
| order by TimeGenerated asc
```

Results | Chart | Add bookmark

TimeGenerated [UTC]	EventID	Computer	Account	SubjectUserName	TargetUserName	Activity
04/10/2026, 12:33:53.749	4625	MyWindowsVM	MicrosoftAccount\soc_user01	-	soc_user01	4625 - An account failed to log on.
04/10/2026, 12:34:01.627	4624	MyWindowsVM	MyWindowsVM\soc_user01	-	soc_user01	4624 - An account was successfully logged on.
04/10/2026, 12:34:04.660	4624	MyWindowsVM	MyWindowsVM\soc_user01	-	soc_user01	4624 - An account was successfully logged on.
04/10/2026, 12:34:16.061	4624	MyWindowsVM	MyWindowsVM\soc_user01	MyWindowsVM$	soc_user01	4624 - An account was successfully logged on.
04/10/2026, 13:26:39.389	4624	MyWindowsVM	MyWindowsVM\soc_user01	-	soc_user01	4624 - An account was successfully logged on.
04/10/2026, 13:26:41.661	4624	MyWindowsVM	MyWindowsVM\soc_user01	-	soc_user01	4624 - An account was successfully logged on.
04/10/2026, 13:26:55.042	4624	MyWindowsVM	MyWindowsVM\soc_user01	MyWindowsVM$	soc_user01	4624 - An account was successfully logged on.
04/10/2026, 13:34:04.772	4625	MyWindowsVM	MicrosoftAccount\soc_user01	-	soc_user01	4625 - An account failed to log on.
04/10/2026, 13:34:09.427	4625	MyWindowsVM	MicrosoftAccount\soc_user01	-	soc_user01	4625 - An account failed to log on.
04/10/2026, 13:34:13.440	4625	MyWindowsVM	MicrosoftAccount\soc_user01	-	soc_user01	4625 - An account failed to log on.
04/10/2026, 13:34:17.582	4625	MyWindowsVM	MicrosoftAccount\soc_user01	-	soc_user01	4625 - An account failed to log on.
04/10/2026, 13:34:22.346	4625	MyWindowsVM	MicrosoftAccount\soc_user01	-	soc_user01	4625 - An account failed to log on.
04/10/2026, 13:34:48.547	4624	MyWindowsVM	MyWindowsVM\soc_user01	-	soc_user01	4624 - An account was successfully logged on.
04/10/2026, 13:34:50.241	4624	MyWindowsVM	MyWindowsVM\soc_user01	-	soc_user01	4624 - An account was successfully logged on.
04/10/2026, 13:34:58.971	4624	MyWindowsVM	MyWindowsVM\soc_user01	MyWindowsVM$	soc_user01	4624 - An account was successfully logged on.
04/10/2026, 14:00:47.275	4625	MyWindowsVM	MicrosoftAccount\soc_user01	-	soc_user01	4625 - An account failed to log on.
04/10/2026, 14:02:45.370	4625	MyWindowsVM	MicrosoftAccount\soc_user02	-	soc_user02	4625 - An account failed to log on.
04/10/2026, 14:07:40.452	4625	MyWindowsVM	MicrosoftAccount\soc_user01	-	soc_user01	4625 - An account failed to log on.

What this query does

- The query looks for events involving your lab accounts whose names contain **soc_**

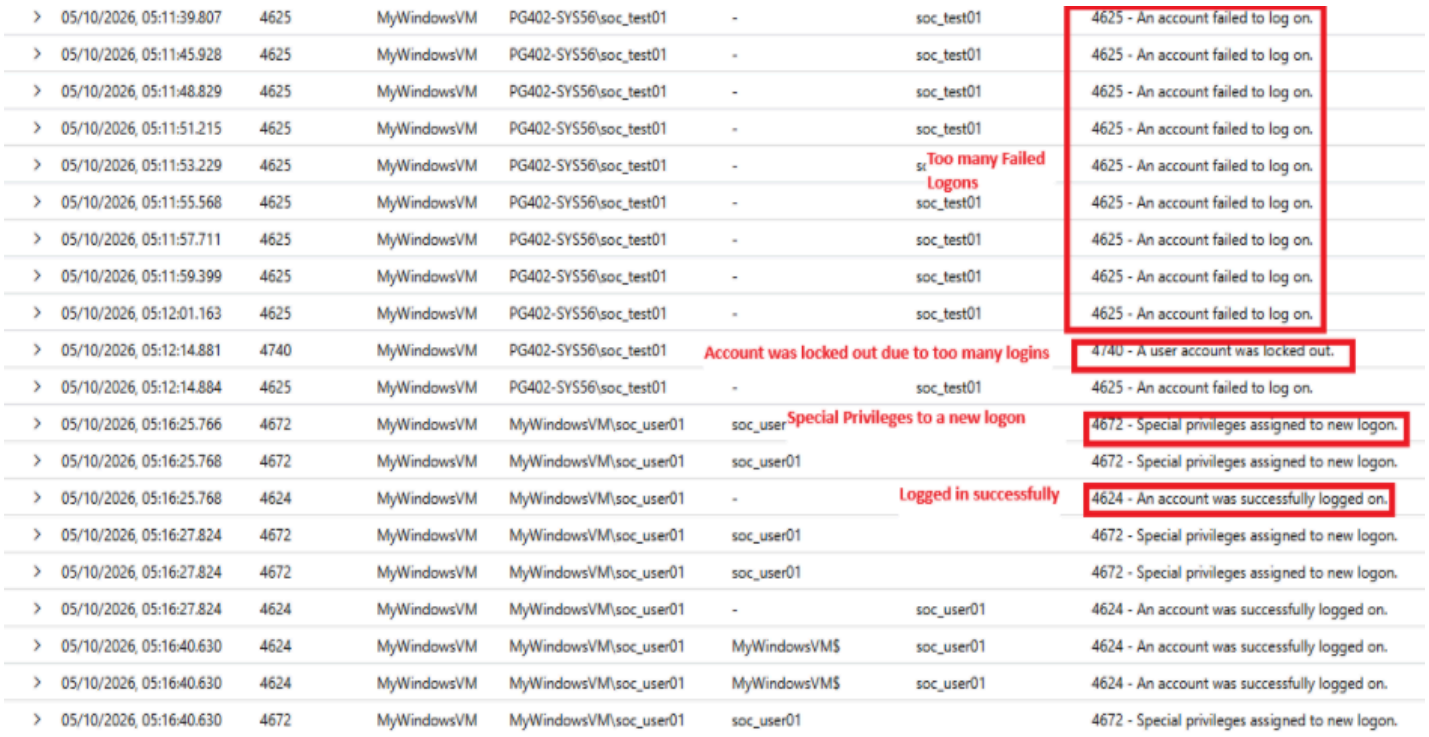
- It then limits the results to important security events:

Event ID	Meaning
4624	Successful logon
4625	Failed logon
4648	Logon using explicit credentials
4672	Special privileges assigned to a logon
4688	Process creation
4697	Service installed
4720	User account created
4728	Member added to a security-enabled global group
4732	Member added to a security-enabled local group
4740	Account locked out
1102	Security audit log cleared

- The query sorts the events using: **| order by TimeGenerated asc**
- This places the oldest event first and newest event last, allowing you to construct a timeline.

3. Construct the Investigation Timeline

- Look at the **TimeGenerated** column.
- Start from the earliest relevant event.
- Record the events in chronological order.
- Pay particular attention to events involving a sequence such as:



>	05/10/2026, 05:11:39.807	4625	MyWindowsVM	PG402-SYS56\soc_test01	-	soc_test01	4625 - An account failed to log on.
>	05/10/2026, 05:11:45.928	4625	MyWindowsVM	PG402-SYS56\soc_test01	-	soc_test01	4625 - An account failed to log on.
>	05/10/2026, 05:11:48.829	4625	MyWindowsVM	PG402-SYS56\soc_test01	-	soc_test01	4625 - An account failed to log on.
>	05/10/2026, 05:11:51.215	4625	MyWindowsVM	PG402-SYS56\soc_test01	-	soc_test01	4625 - An account failed to log on.
>	05/10/2026, 05:11:53.229	4625	MyWindowsVM	PG402-SYS56\soc_test01	-	s	4625 - An account failed to log on.
>	05/10/2026, 05:11:55.568	4625	MyWindowsVM	PG402-SYS56\soc_test01	-	soc_test01	4625 - An account failed to log on.
>	05/10/2026, 05:11:57.711	4625	MyWindowsVM	PG402-SYS56\soc_test01	-	soc_test01	4625 - An account failed to log on.
>	05/10/2026, 05:11:59.399	4625	MyWindowsVM	PG402-SYS56\soc_test01	-	soc_test01	4625 - An account failed to log on.
>	05/10/2026, 05:12:01.163	4625	MyWindowsVM	PG402-SYS56\soc_test01	-	soc_test01	4625 - An account failed to log on.
>	05/10/2026, 05:12:14.881	4740	MyWindowsVM	PG402-SYS56\soc_test01			4740 - A user account was locked out.
>	05/10/2026, 05:12:14.884	4625	MyWindowsVM	PG402-SYS56\soc_test01	-	soc_test01	4625 - An account failed to log on.
>	05/10/2026, 05:16:25.766	4672	MyWindowsVM	MyWindowsVM\soc_user01	soc_user		4672 - Special privileges assigned to new logon.
>	05/10/2026, 05:16:25.768	4672	MyWindowsVM	MyWindowsVM\soc_user01	soc_user01		4672 - Special privileges assigned to new logon.
>	05/10/2026, 05:16:25.768	4624	MyWindowsVM	MyWindowsVM\soc_user01	-		4624 - An account was successfully logged on.
>	05/10/2026, 05:16:27.824	4672	MyWindowsVM	MyWindowsVM\soc_user01	soc_user01		4672 - Special privileges assigned to new logon.
>	05/10/2026, 05:16:27.824	4672	MyWindowsVM	MyWindowsVM\soc_user01	soc_user01		4672 - Special privileges assigned to new logon.
>	05/10/2026, 05:16:27.824	4624	MyWindowsVM	MyWindowsVM\soc_user01	-	soc_user01	4624 - An account was successfully logged on.
>	05/10/2026, 05:16:40.630	4624	MyWindowsVM	MyWindowsVM\soc_user01	MyWindowsVM$	soc_user01	4624 - An account was successfully logged on.
>	05/10/2026, 05:16:40.630	4624	MyWindowsVM	MyWindowsVM\soc_user01	MyWindowsVM$	soc_user01	4624 - An account was successfully logged on.
>	05/10/2026, 05:16:40.630	4672	MyWindowsVM	MyWindowsVM\soc_user01	soc_user01		4672 - Special privileges assigned to new logon.

- The lab's intended investigation storyline is to correlate events rather than interpret one event in isolation.

X. Attack → Evidence → Detection

Give students this table as their lab worksheet.

Simulation	Primary Event	What Students Investigate
Failed login	**4625**	Account, source IP, frequency
Successful login	**4624**	User, source, logon type
Password spray	**4625**	Number of targeted accounts
Account creation	**4720**	Creator and new account
Admin-group modification	**4732**	Who added whom
Privileged logon	**4672**	Privileges and account

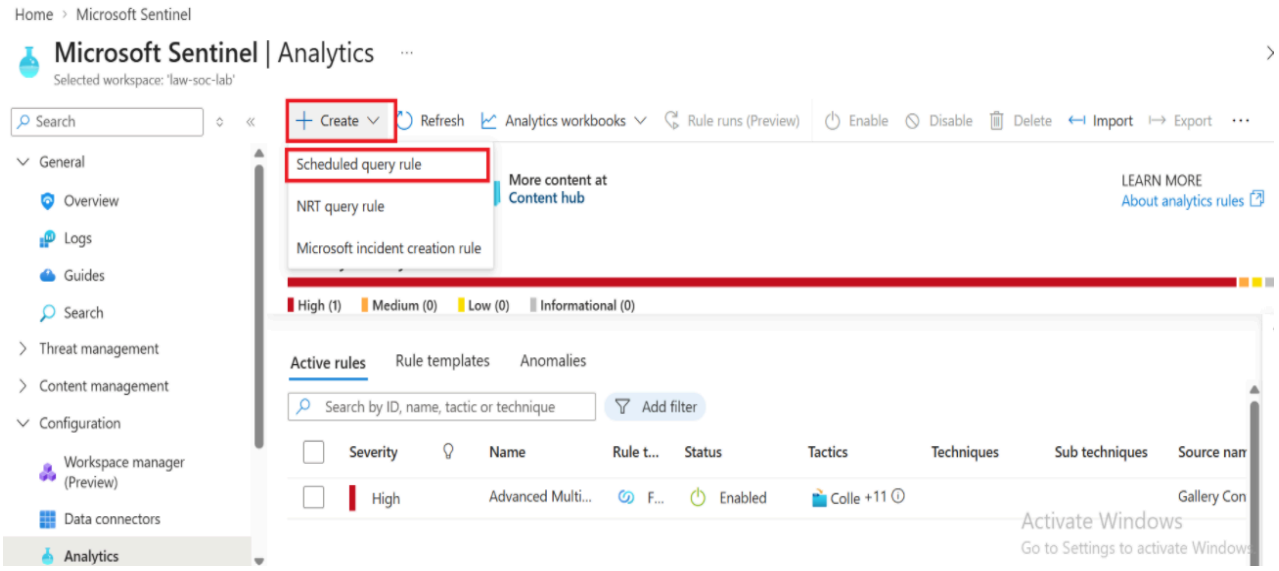
Process execution	**4688 / Sysmon 1**	Process + parent + command line
Network activity	**Sysmon 3**	Destination + process
File creation	**Sysmon 11**	File + creating process
DNS	**Sysmon 22**	Domain + process
Registry change	**Sysmon 12–14**	Key/value + process
Account lockout	**4740**	Account + source
Scheduled task	**4698**	Task + creator
Log clearing	**1102**	Who cleared the log

Important SOC Principle

Sysmon's event model is designed to provide detailed behavioral telemetry. However, **Sysmon events are evidence rather than automatic alerts**. SOC analysts must use **context and correlation** to determine whether activity is suspicious.

XI. Create Sentinel Analytics Rules

- Open the Azure portal.
- Open Microsoft Sentinel.
- Select your workspace, `law-soc-lab`.
- From the left menu, select **Analytics** from the **Configuration** menu**.**
- It shows a message *"This page was moved to Defender portal, please connect your workspace to the Defender portal"*. **Just ignore this warning and refresh the analytics page by clicking Ctrl+R.**
- Click on **Create** → **Scheduled query** rule.



Rule 1 - Multiple Failed Logins

- Enter the following details:
  - Rule Name: **SOC Lab - Multiple Failed Logins**
  - Description: **Detects repeated failed login attempts on a Windows account.**
  - Severity: **Medium**
  - Status: **Enabled**
- Click **Next.**
- Under Rule query, paste the following KQL query:

```
SecurityEvent

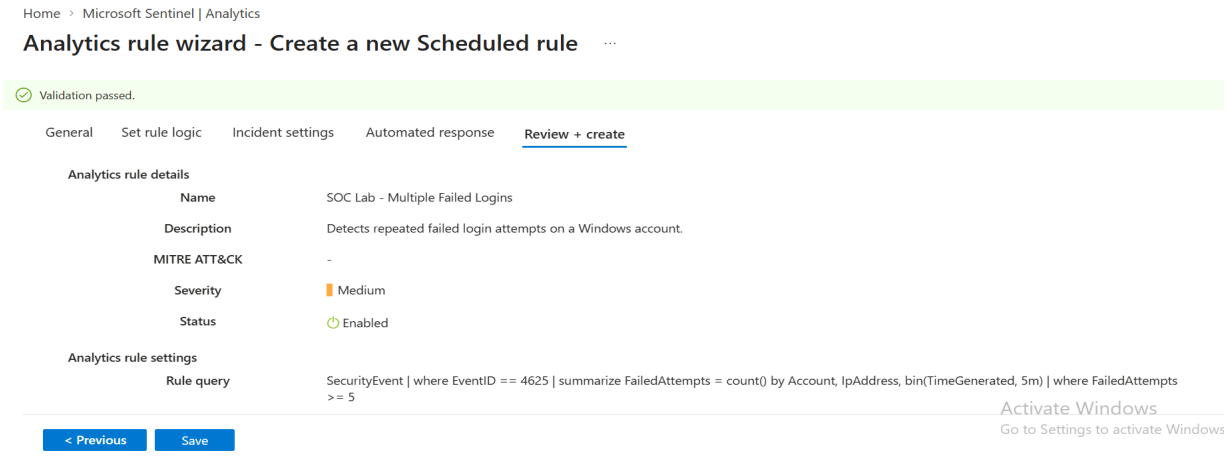
| where EventID == 4625

| summarize FailedAttempts = count()

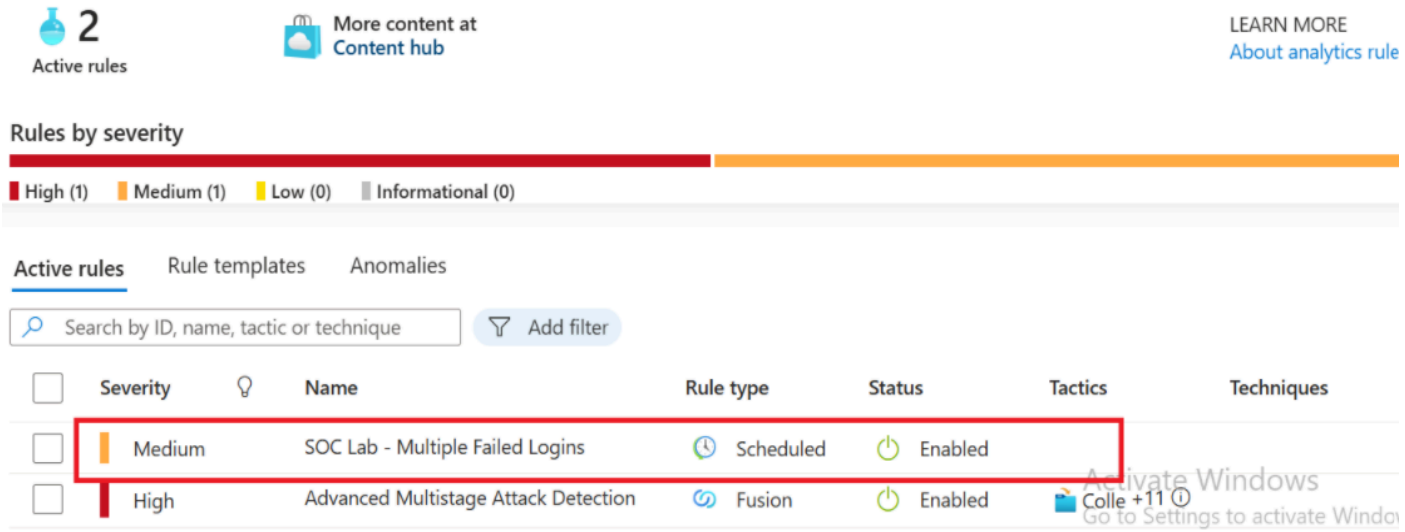
        by Account, IpAddress, bin(TimeGenerated, 5m)

| where FailedAttempts >= 5
```

- Under query scheduling:
  - Run query every: **5 minutes**
  - Lookup data from the last: **5 minutes**
- Under Alert threshold, set the rule to trigger when the number of results is **greater than 0.**
- Keep the remaining settings at their defaults. Click on **Review+Create.** Then click **Save.**



- After creation, the rule **SOC Lab - Multiple Failed Logins** is displayed like this:



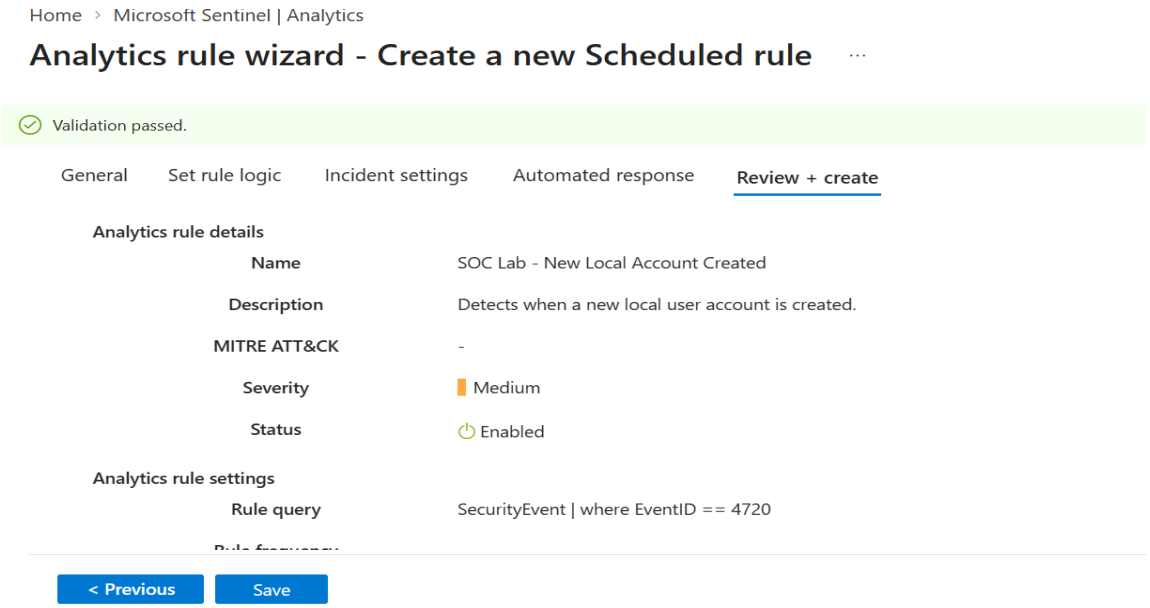
Rule 2 - New Local Account Created

- Enter the following details:
  - Rule Name: **SOC Lab - New Local Account Created**
  - Description: **Detects when a new local user account is created.**
  - Severity: **Medium**
  - Status: **Enabled**
- Click **Next**.
- Under Rule query, paste the following KQL query:

  SecurityEvent

  | where EventID == 4720

- Under query scheduling:
  - Run query every: **5 minutes**
  - Lookup data from the last: **5 minutes**
- Under Alert threshold, set the rule to trigger when the number of results is **greater than `0`.**
- Keep the remaining settings at their defaults. Click on **Review+Create.** Then click **Save.**



Rule 3 - User Added to Administrators

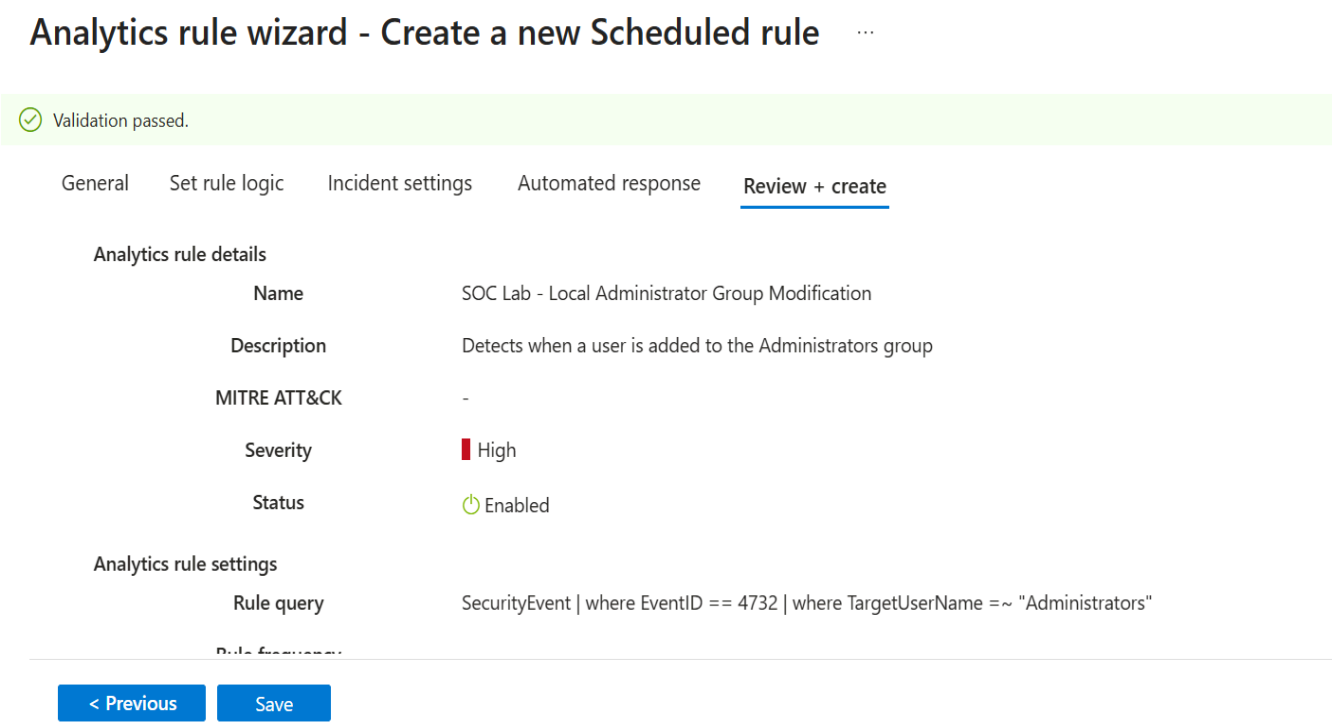
- Enter the following details:
  - Rule Name: **SOC Lab - Local Administrator Group Modification**
  - Description: **Detects when a user is added to the Administrators group.**
  - Severity: **High**
  - Status: **Enabled**
- Click **Next.**
- Under Rule query, paste the following KQL query:

```
SecurityEvent

| where EventID == 4732

| where TargetUserName =~ "Administrators"
```

- Under query scheduling:
  - Run query every: **5 minutes**
  - Lookup data from the last: **5 minutes**
- Under Alert threshold, set the rule to trigger when the number of results is **greater than `0`.**
- Keep the remaining settings at their defaults. Click on **Review+Create.** Then click **Save.**



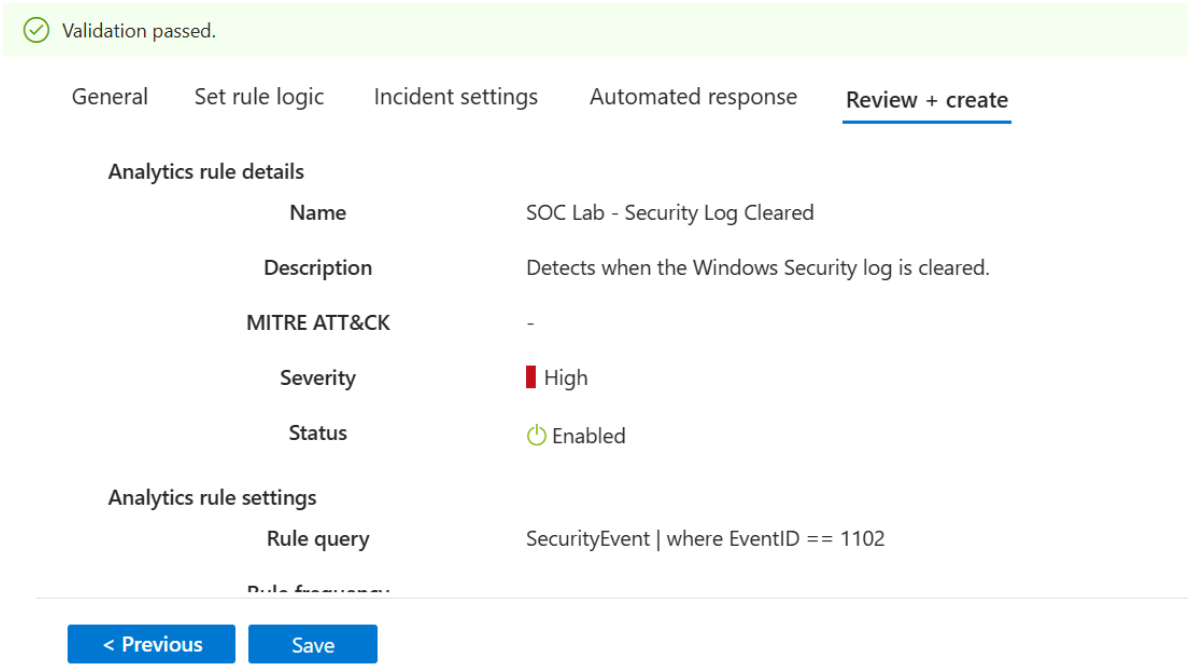
Rule 4 - Security Log Cleared

- Enter the following details:
  - Rule Name: **SOC Lab - Security Log Cleared**
  - Description: **Detects when the Windows Security log is cleared.**
  - Severity: **High**
  - Status: **Enabled**
- Click Next: Set rule logic.
- Under Rule query, paste the following KQL query:

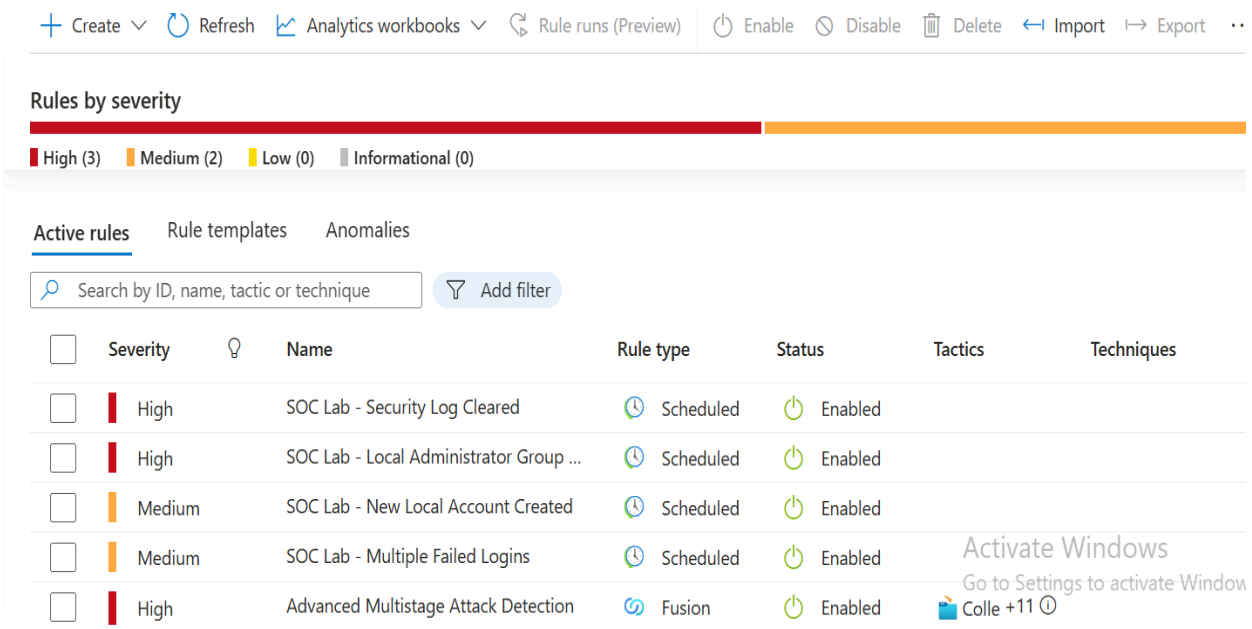
```
SecurityEvent

| where EventID == 1102
```

- Under query scheduling:
  - Run query every: **5 minutes**
  - Lookup data from the last: **5 minutes**
- Under Alert threshold, set the rule to trigger when the number of results is **greater than `0`.**
- Keep the remaining settings at their defaults. Click on **Review+Create.** Then click **Save.**



- After creating all the analytical rules, they are displayed like this:



XII. Review Questions and Analysis

1a. Identify and explain the purpose and availability of the Microsoft Sentinel, Log Analytics Workspace, Azure Monitor Agent (AMA), Data Collection Rule, and KQL tools used in the SOC attack simulation. Explain how each tool supports security monitoring and ethical security investigation. (CO1)

Answer:

The following tools work together to collect, analyze, and investigate security events in a SOC environment.

- **Microsoft Sentinel:** A cloud-based SIEM solution that detects threats, generates alerts, investigates incidents, and supports security response.
- **Log Analytics Workspace:** Stores collected security logs and allows analysts to search and investigate events.
- **Azure Monitor Agent (AMA):** Collects security events from the Windows virtual machine and sends them to the Log Analytics Workspace.
- **Data Collection Rule (DCR):** Specifies which events should be collected and where the collected data should be sent.
- **Kusto Query Language (KQL):** Queries and analyzes security logs to identify suspicious activities, such as failed logins and unauthorized account changes.

How they support security monitoring: AMA collects events according to the DCR, and the events are stored in the Log Analytics Workspace. Analysts use KQL to investigate the data, while Microsoft Sentinel uses detection rules to generate alerts.

Ethical investigation: These tools help students simulate attacks in an authorized, isolated lab, examine the resulting events, and understand how security incidents are detected without affecting real users or systems.

1b. From the simulated authentication failures, password-spray activity, account creation, and privilege modification events, discuss the business and security implications of these activities and explain how Microsoft Sentinel can support security policies for protecting organizational accounts and resources. (CO2)

Answer:

The simulated activities can indicate attempts to gain unauthorized access or misuse system privileges.

- **Authentication failures:** Repeated failed logins may indicate password guessing or brute-force attempts.
- **Password spraying:** An attacker tries common passwords against multiple accounts to avoid triggering account-specific lockout thresholds.
- **Account creation:** Unauthorized accounts may allow attackers to maintain access to a system.
- **Privilege modification:** Adding an unauthorized user to an administrator group may give them access to sensitive data and system settings.

Business and security implications: These activities can cause data theft, financial loss, service disruption, privacy violations, and reputational damage. Unauthorized administrator access can also allow attackers to change security settings or create additional accounts.

Role of Microsoft Sentinel:

- Detects repeated failed logins and suspicious account activity.
- Generates alerts for account creation and privilege changes.
- Helps analysts investigate related events and identify possible attacks.
- Supports security policies such as strong passwords, multi-factor authentication (MFA), least privilege, and regular account monitoring.

Thus, Microsoft Sentinel helps organizations identify potential threats and respond before further damage occurs.

1c. Develop suitable detection logic and investigation metrics in Microsoft Sentinel for the simulated security activities. Explain how the selected policies, thresholds, and correlation rules help distinguish normal activity from potentially malicious behavior. (CO3)

Answer:

Microsoft Sentinel uses KQL detection queries to identify suspicious activities in collected Windows security logs.

1. Multiple failed logins — Event ID 4625

```
SecurityEvent
| where EventID == 4625
| summarize FailedAttempts = count()
    by Account, IpAddress, bin(TimeGenerated, 5m)
| where FailedAttempts >= 5
```

This query detects five or more failed login attempts for the same account and IP address within a five-minute interval.

2. New account creation — Event ID 4720

```
SecurityEvent
| where EventID == 4720
```

This identifies events associated with new user account creation.

3. User added to Administrators — Event ID 4732

```
SecurityEvent
| where EventID == 4732
| where TargetUserName =~ "Administrators"
```

This identifies additions to the local Administrators group when the recorded group name matches.

4. Security log cleared — Event ID 1102

```
SecurityEvent
| where EventID == 1102
```

This detects events indicating that the Windows Security audit log was cleared.

Investigation metrics:

- Number and frequency of failed logins.

- Source IP address and targeted account.
- Time of account creation.
- Users added to administrator groups.
- Time at which the Security log was cleared.

Role of thresholds and correlation:

A threshold of five failed logins within five minutes helps identify unusual authentication activity. Correlating events using timestamps, account names, host names, and source IP addresses helps analysts identify related activities.

These rules must be adjusted to the organization's normal activity. Legitimate users may also generate failed logins, and legitimate administrators may create accounts. Therefore, alerts require investigation before being classified as malicious.

1d. Analyze the security risks and potential dangers associated with conducting controlled penetration-testing activities such as password spraying, account creation, privilege modification, process execution, and network/DNS activity, and explain the precautions required to perform them safely in an isolated environment. (CO4)

Answer:

Controlled penetration testing helps identify security weaknesses, but improper testing can affect system availability, confidentiality, and integrity.

Security risks:

1. Password spraying: May cause account lockouts or unauthorized access if a password is guessed correctly.
2. Account creation: Unauthorized accounts may provide attackers with continued access.
3. Privilege modification: Excessive privileges may expose sensitive information or allow unwanted system changes.
4. Process execution: Unknown or malicious programs may damage files, alter configurations, or compromise the VM.
5. Network and DNS activity: Excessive traffic or incorrect configurations may disrupt connectivity or affect systems outside the lab.

Precautions for safe testing:

- Obtain authorization before performing tests.
- Use isolated virtual machines and a dedicated lab network.
- Use test accounts and non-sensitive data.
- Take VM snapshots before testing so the system can be restored.
- Limit authentication attempts and network traffic.
- Monitor security logs during the exercises.
- Avoid targeting public systems or devices outside the authorized environment.
- Restore the lab to a clean state after testing.

These precautions reduce the risk of data loss, service disruption, and unauthorized access while ensuring that testing remains safe and ethical.

1e. Interpret the security telemetry generated during the simulation using KQL. Correlate the events to construct an incident timeline, identify the affected host and account, determine the likely attack

sequence, and recommend an appropriate response action based on the investigation findings. (CO5)

Answer:

Security telemetry from the Windows VM can be investigated using KQL in the Log Analytics Workspace.

1. Query security events

```
SecurityEvent
| where EventID in (4624, 4625, 4720, 4732, 1102)
| project TimeGenerated, EventID, Computer,
        Account, SubjectUserName, TargetUserName, Activity
| order by TimeGenerated asc
```

This query retrieves selected login, account creation, administrator group modification, and Security log-clearing events in chronological order.

2. Construct an incident timeline

The analyst examines event timestamps to understand the sequence of activities. For example:

- Repeated failed logins may indicate password guessing.
- A successful login shortly afterward may require investigation.
- A new account or administrator group modification may indicate an attempt to establish access or increase privileges.
- A cleared Security log may indicate an attempt to remove evidence.

This is a possible attack sequence, not proof that the events are connected.

3. Identify the affected host and account

The `Computer` field identifies the host that recorded the event. Fields such as `Account`, `SubjectUserName`, and `TargetUserName` help identify the initiating user and the account affected. The available fields vary by event type.

4. Correlate the events

Compare timestamps, host names, account names, and source IP addresses where available. Check whether multiple suspicious events involve the same host or account. This helps determine whether the events may be part of a single incident.

5. Recommend response actions

- Investigate suspicious login attempts and source IP addresses.
- Reset compromised credentials and revoke active sessions when necessary.
- Disable unauthorized accounts.
- Remove unauthorized administrator privileges.
- Preserve relevant logs and investigate suspicious processes.
- Document findings and improve detection rules.

Conclusion: KQL-based investigation helps analysts build an incident timeline, identify affected systems and accounts, assess possible attack activity, and recommend suitable response actions.